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A Probabilistic Gaia DR3 Catalogue of Stars with Very Low Galactic Rest-Frame Speeds

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ABSTRACT

We present a Gaia DR3 catalogue of stars selected by very low Galactic rest-frame speed, $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$. Using Lindegren et al. (2021) parallax zero-point corrections, Bailer-Jones et al. (2021) photogeometric distances, and Monte Carlo propagation of the Gaia parallax–proper-motion covariance, we identify a headline Tier A+B sample of 517 stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$ and a broader Tier A+B+C sample of 1,835 stars with $P > 0.50$. Orbit integrations in the static, axisymmetrized Hunter et al. (2024) Galactic potential map the broader sample onto compact, highly radial model orbits, with median eccentricity 0.949, median pericentre 151 pc, and median apocentre 7.45 kpc. The observed Tier A+B+C count exceeds the expectation from a control-matched smooth velocity distribution by a factor of 24, and Anderson–Darling tests reject continuity with the adjacent 25–50 $\rm km\,s^{-1}$ band ($p < 0.001$). A controlled GeDR3mock injection-recovery exercise recovers 60.9% of true sub-25 $\rm km\,s^{-1}$ stars end-to-end (87.6% conditional on entering the pipeline). The dominant selection-function systematic is the 83.8% of headline-catalogue stars that fall in prior-dominated, low-parent-count ($n < 10$) GaiaUnlimited cells, which anchor an effective-volume inflation factor of 2.13. We release the catalogue as a probabilistic kinematic sample with model-dependent orbit diagnostics and subset-qualified chemical context.

Keywords: Milky Way Galaxy (1054) — Milky Way dynamics (1051) — Galaxy structure (622) — Stellar kinematics (1608) — Astrometry (80) — Sky surveys (1464) — Catalogs (205)

1. INTRODUCTION

Gaia DR3 provides full six-dimensional phase-space information for tens of millions of stars through its astrometry and radial velocities (Gaia Collaboration et al. 2023; Katz et al. 2023). This makes it possible to search the observed Milky Way phase-space distribution for rare kinematic states, not only for previously defined stellar populations or merger debris. Here we focus on one such state: stars observed with unusually low instantaneous Galactic rest-frame speed. The broader Galactic-structure and Gaia-era archaeology context is reviewed by Binney & Merrifield (1998), Bland-Hawthorn & Gerhard (2016), Helmi (2020), and Deason & Belokurov (2024). This complements Gaia-era catalogues selected at the opposite kinematic extreme, including Gaia DR3 hypervelocity-star searches (Marchetti et al. 2022; Carballo-Bello et al. 2025), and halo catalogues selected from high tangential velocity or reduced proper motion (Viswanathan et al. 2023). The closest published Gaia-era catalogue selected by low Galactocentric azimuthal velocity is Filion et al. (2025), which identifies ~ 70 slow- and retrograde stars with thin-disc-like abundances. That work focuses on chemodynamic outliers within a disc-like chemistry box and is therefore complementary to the chemistry-agnostic, low-instantaneous-speed catalogue presented here.

The sample is defined by present-day kinematics, not by inferred origin. We construct a probabilistic low-$V_{\rm grf}$ Gaia DR3 catalogue, quantify the stability of the velocity-threshold membership under measurement uncertainties, and report model-dependent orbit diagnostics in an adopted Milky Way potential. The orbit integrations are used to interpret the observed slow state, but they do not define the catalogue.

This framing is important because low instantaneous speed can arise in more than one physical context. For wide radial orbits it may mark an apocentric phase, as in the apocentre-pile-up interpretation of radial halo debris (Deason et al. 2018); for compact inner-Galaxy orbits it

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may reflect a different part of the orbit distribution. We therefore compare the low-V_{grf} catalogue with matched Gaia DR3 control samples across higher velocity bands and treat external spectroscopic matches as context for the subset with available chemistry, not as a whole-catalogue population assignment.

Section 2 describes the sample definition, uncertainty propagation, controls, and adopted orbit model. Section 3 presents the empirical catalogue properties and chemical cross-matches. Section 4 gives the model-dependent dynamical results. Section 5 discusses the interpretation and its limits, and Section 6 summarises the conclusions.

2. SAMPLE DEFINITION AND METHODS

2.1. *Gaia DR3 Sample*

The parent source is Gaia DR3 (Gaia Collaboration et al. 2023), restricted to stars with full six-dimensional phase-space information: sky position, parallax, proper motion, and radial velocity. Radial velocities in DR3 are available for approximately 33.8 million stars observed by the Radial Velocity Spectrometer (Katz et al. 2023). This 6D subset has a non-trivial selection function in sky position, magnitude, and colour (Castro-Ginard et al. 2023; Cantat-Gaudin et al. 2023); it is not a neutral slice of the Galaxy, and distant 6D samples are commonly giant-heavy, with incompleteness expected for cool faint dwarfs.

The sample was constructed from a local mirror of the Gaia DR3 source table using the columns needed for a six-dimensional solution: `source_id`, `ra`, `dec`, `parallax`, `pmra`, `pmdec`, `radial_velocity`, the corresponding uncertainties and astrometric correlations, broad-band photometry, `ruwe`, and `duplicated_source`. The equivalent Gaia Archive selection is `radial_velocity IS NOT NULL`, `parallax > 0`, `pmra IS NOT NULL`, `pmdec IS NOT NULL`, and `parallax_over_error > 5`. No RUWE or duplicated-source cut is applied to define the point-estimate catalogue; those flags define the Gold subset below. Heliocentric observables are transformed into Galactocentric coordinates and velocities, and Galactic rest-frame speeds are computed.

To avoid imposing the final catalogue boundary during candidate construction, we first scanned the full local Gaia DR3 source mirror and retained an intentionally broad legacy buffer with preliminary $V_{\mathrm{grf}} < 200\,\mathrm{km\,s^{-1}}$. This buffer contains 5,867,654 unique Gaia source IDs. After the final parallax zero-point, photogeometric-distance, and Solar-frame conventions are applied, 2,591 sources have corrected point-estimate $V_{\mathrm{grf}} < 25\,\mathrm{km\,s^{-1}}$ within the 20,829-source near-threshold candidate pool.

Table 1. Sample accounting under the final distance and Solar-frame conventions. The candidate pool is the Gaia DR3 6D source list recovered from the full-parent buffered scan; the corrected point-estimate row applies the Lindegren+2021 parallax zero-point, Bailer-Jones+2021 photogeometric distances, and the default Galactocentric frame before evaluating the $V_{\mathrm{grf}} <$ 25 $\mathrm{km\,s^{-1}}$ boundary. Tiers are mutually exclusive except for the explicitly cumulative Tier A+B and Tier A+B+C rows.

Set	N
Gaia DR3 6D candidate pool	20,829
Corrected point-estimate $V_{\mathrm{grf}} < 25\ \mathrm{km\,s^{-1}}$	2,591
Tier A, $P(V_{\mathrm{grf}} < 25) > 0.95$	276
Tier B increment, $0.84 < P(V_{\mathrm{grf}} < 25) \le 0.95$	241
Tier A+B, $P(V_{\mathrm{grf}} < 25) > 0.84$	517
Tier C increment, $0.50 < P(V_{\mathrm{grf}} < 25) \le 0.84$	1,318
Tier A+B+C, $P(V_{\mathrm{grf}} < 25) > 0.50$	1,835
Tier D, point-estimate $V_{\mathrm{grf}} < 25\ \mathrm{km\,s^{-1}}$ and $P \le 0.50$	756
Candidate-pool sources outside the corrected cut	18,238

Monte Carlo propagation of the observables then defines the catalogue tiers used throughout the paper: Tier A ($P > 0.95$, 276 stars), Tier A+B ($P > 0.84$, 517 stars), and Tier A+B+C ($P > 0.50$, 1,835 stars). The remaining 756 corrected point-estimate sources below $25\,\mathrm{km\,s^{-1}}$ have $P \le 0.50$ and are retained as Tier D rather than as part of the headline catalogue.

This is not a volume-complete sample and it is not intended as a complete inventory of all such stars in the Milky Way. It is a Gaia DR3 6D catalogue defined by a simple instantaneous kinematic state and reported with explicit threshold-membership probabilities.

To keep the role of each count unambiguous, Table 2 maps each tier and derived sample to the results it anchors. The **headline catalogue** for the paper is Tier A+B ($N = 517$); orbit-derived medians and population distributions are computed on the broader Tier A+B+C sample ($N = 1{,}835$). Selection-function and chemistry diagnostics are reported only as population-scale context, never as a redefinition of the headline.

2.2. *Quality Control*

Because the slowest stars are the most vulnerable to measurement error, we report conservative high-quality

Table 2. Which sample anchors which result. The headline catalogue for the paper is Tier A+B ($N = 517$); orbit medians and population distributions use Tier A+B+C (N = 1,835). Every numerical claim in the paper names its anchoring sample explicitly.

Sample	Used for
Tier A ($N = 276$, $P > 0.95$)	Highest-confidence empirical subset; robustness anchor.
Tier A+B ($N = 517$, $P > 0.84$)	**Headline catalogue**; Gold subsets; selection-function context.
Tier A+B+C (N = 1,835, $P > 0.50$)	Orbit-derived medians; broad catalogue characterisation.
α-classified ($N = 113$ in Tier A+B+C)	Splash/GSE/Aurora/disc fractions only.
Tier A+B SF-weighted	Population-scale context; never per-star weights.

Table 3. Sample rows from the Tier A+B headline catalogue. The full table is available in machine-readable form in the review package as `machine_readable_tables/catalogue_tierAB_mrt.txt`; the FITS version is `catalogues/catalogue_expanded_tierAB.fits`.

Gaia DR3 source_id	R.A. (deg)	Decl. (deg)	$V_{\rm grf}$ (km s^{-1})	$P(V_{\rm grf} < 25)$	$N_{\rm MC}$	Tier	RVS QC
10283049355584128	50.09688	6.77727	16.29	0.900	500	B	ok
19320583963797632	39.10617	7.09534	19.74	0.926	500	B	ok
36729808698742528	57.56126	11.89217	20.12	0.960	500	A	ok
47180254402996480	61.88618	18.71729	15.45	0.972	500	A	ok
77358206452039808	32.47388	14.49229	13.16	0.956	500	A	ok
78171462803918464	29.93784	15.14448	11.64	0.944	500	B	ok
127624407740600576	37.46443	27.57781	9.93	0.878	500	B	ok
197752599391374208	89.19567	46.29982	16.39	0.852	500	B	ok

Note—Only eight rows are shown for form and content. The electronic table contains all 517 Tier A+B sources.

subsets in addition to the tiered catalogue. The Gold observational-quality subset is defined inside Tier A+B by RUWE < 1.4, an acceptable DR3 RVS quality flag, parallax signal-to-noise > 10, and fractional distance uncertainty $\sigma_d/d < 0.15$. Dynamical robustness is reported separately in Section 4.5, where additional orbit-derived criteria are applied.

Several later diagnostics necessarily use narrower subsets. The α-classified spectroscopic subset is defined only by the availability of APOGEE or GALAH [Fe/H] and alpha-element abundances within Tier A+B+C; it is used for within-subset literature-region comparisons, not for whole-catalogue population fractions. The selection-function inverse-weighted sums are likewise reported only as observability context, because many slow stars occupy sparse colour–magnitude cells where the Gaia DR3 RVS selection-function tool returns its prior mean.

Table 4. Sensitivity of the main orbit summaries to removing stars flagged as poor by the internal Gaia DR3 RVS-quality screen. Values are point-estimate QA diagnostics; the qualitative compact, radial-orbit result is unchanged.

Sample	N	$\widetilde{R_{\rm peri}}$ (pc)	$\tilde{e}$
Tier A+B+C, all RV qualities	1,835	112	0.964
Tier A+B+C, excluding poor RVS	1,743	113	0.964
Tier A+B, all RV qualities	517	112	0.973
Tier A+B, excluding poor RVS	482	114	0.973

2.3. *Velocity-Threshold Uncertainty*

The catalogue is defined by threshold-membership probability rather than by a hard point estimate alone. For every source in the propagated candidate pool we draw Monte Carlo realisations of $(\varpi, \mu_{\alpha*}, \mu_\delta)$ from the Gaia parallax–proper-motion covariance submatrix, holding sky position fixed because the positional uncertainties are negligible for the velocity scale considered here. We draw radial velocity independently from its quoted uncertainty, apply the Lindegren et al. (2021) zero-point correction, and draw distance from the asymmetric Bailer-Jones et al. (2021) photogeometric posterior, following the parallax-handling guidance of Luri et al. (2018). We then recompute $V_{\rm grf}$ for each realisation using the same Solar parameters and coordinate convention as the point-estimate catalogue. This explicit propagation is complementary to recent Gaia DR3 Cartesian catalogue work that compares second-order and Monte Carlo covariance propagation for the full RVS sample (Zhang et al. 2025).

Tier-fraction percentages reported below carry $\mathrm{Beta}(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011); integer-count diagnostics (e.g., bridger counts) carry Gehrels-style Poisson 1σ bands.

The Monte Carlo scheme uses 500 baseline realisations per source, refines stars in the transition band $0.30 < P < 0.70$ to 5,000 realisations, and assigns 10,000 realisations to stars whose point-estimate $V_{\rm grf}$ lies within $2\,\mathrm{km\,s^{-1}}$ of the threshold. A separate 100-star convergence check compares 500 and 5,000 realisations. We use the resulting $P(V_{\rm grf} < 25\,\mathrm{km\,s^{-1}})$ values to define Tier A ($P > 0.95$), Tier B increment ($0.84 < P \leq 0.95$), Tier C increment ($0.50 < P \leq 0.84$), and Tier D (corrected point estimate below $25\,\mathrm{km\,s^{-1}}$ but $P \leq 0.50$).

The Tier A+B boundary is used as the headline because $P > 0.84$ corresponds approximately to a one-sided one-sigma probability threshold: these stars are more likely than not to remain below $25\,\mathrm{km\,s^{-1}}$ even after a one-sigma upward fluctuation in the propagated threshold variable. The three operating thresholds correspond to standard one-sided Gaussian-tail levels: $P > 0.95$ (Tier A) is approximately a one-sided 1.6σ threshold, $P > 0.84$ (Tier A+B) is approximately a one-sided 1σ threshold, and $P > 0.50$ (Tier A+B+C) is the more-likely-than-not boundary. Tier C is reported as the broader, lower-confidence catalogue extension rather than as part of the conservative headline sample.

Because the Bailer-Jones et al. (2021) photogeometric posterior is itself conditioned on Gaia astrometry, drawing the distance independently of the astrometric covariance is an approximation. The catalogue therefore uses this Monte Carlo draw only to define threshold-membership probabilities, and treats distance-method and covariance-coupling tests as sensitivity diagnostics rather than as part of the catalogue definition.

We quantified this approximation with a 200-star stress test drawn from the transition band. The test keeps the BJ split-normal distance marginal but imposes a conservative Gaussian-copula anti-correlation between the parallax residual and the distance quantile. Relative to the released independent-distance catalogue probabilities, the median $|\Delta P|$ is 0.0061, the 84th percentile is 0.0130, and the largest shift is 0.0274; four of the 200 test stars change tier labels (Fig. 1).

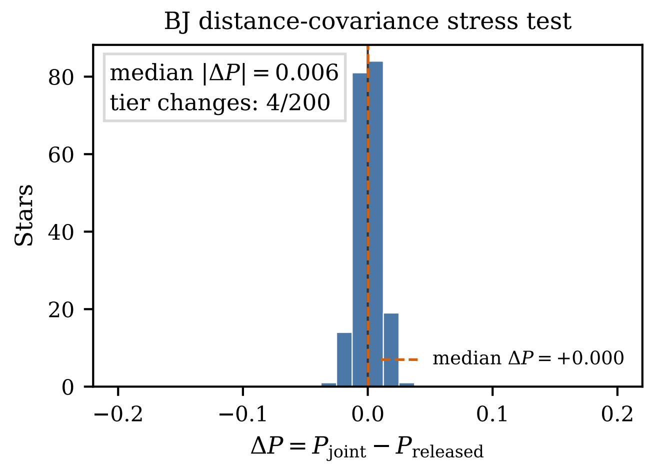

Figure 1. Change in $P(V_{\rm grf} < 25\,\mathrm{km\,s^{-1}})$ when the BJ-style distance draw is coupled to the Gaia parallax residual for 200 transition-band sources. The stress test is deliberately conservative and probes the distance covariance approximation in the catalogue Monte Carlo.

A separate reliability reconstruction repeats the 500-realisation baseline pass with an independent random seed, selects the 2,275 sources with baseline $0.30 < P < 0.70$, and compares deciles of the baseline probability against a held-out 5,000-realisation refinement. The decile medians lie close to the one-to-one relation: the median absolute decile residual is 0.0019 and the largest is 0.0040 (Fig. 2). This diagram calibrates the catalogue tier probabilities themselves; it is distinct from the Gaia RVS selection-function systematic quantified separately in Section 5.

2.4. *Galactocentric Transformation and Supporting Orbit Model*

For the Galactocentric transformation we adopt $R_\odot = 8.178\,\mathrm{kpc}$ (GRAVITY Collaboration et al. 2019), a Solar height above the plane of 25 pc, Solar peculiar motion $(U, V, W) = (11.1, 12.24, 7.25)\,\mathrm{km\,s^{-1}}$ from Schönrich et al. (2010), and $V_{\rm circ} = 232.0\,\mathrm{km\,s^{-1}}$ at the Solar radius, close to recent rotation-curve determinations (Eilers et al. 2019). The coordinate transformation is performed with `astropy.coordinates` (Astropy Collaboration et al. 2013) and the resulting Cartesian Galactocentric phase-space vectors are passed to `agama`. Three additional Solar-parameter variants from the recent literature (GRAVITY Collaboration 2022;

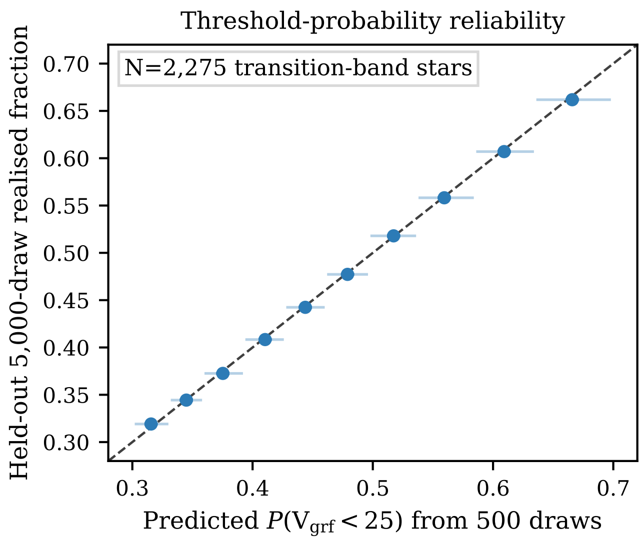

Figure 2. Reliability of the threshold-membership probabilities in the transition band. Points bin stars by the recomputed 500-realisation baseline probability; the ordinate is the realised fraction in an independent held-out 5,000-realisation refinement. Error bars are central 68% Beta$(k + \frac{1}{2}, n - k + \frac{1}{2})$ intervals (Cameron 2011).

Schönrich 2012; Reid & Brunthaler 2020) are used as a sensitivity bracket in Section 4.5.

We apply the per-star Lindegren et al. (2021) Gaia parallax zero-point correction and use the Bailer-Jones et al. (2021) photogeometric distance posterior as the default distance estimator. The inverse-parallax distance, with and without the zero-point correction, is retained only as a sensitivity diagnostic. In the propagated candidate pool, the adopted photogeometric-distance convention gives 2,591 point-estimate sources below the cut before probability-tiering.

The default supporting orbit model is the static, axisymmetrized form of the analytic Milky Way potential of Hunter et al. (2024). The underlying Hunter+2024 model is not Gaia-DR3-calibrated; it combines analytic bulge/bar, nuclear, disc, and halo components constrained by rotation-curve and gas-terminal-velocity data. We use its axisymmetrized form as the catalogue default because it keeps orbital quantities such as E, L_z, and Stäckel-fudge actions well-defined. Orbit integration and action estimates use `agama` (Vasiliev 2019); actions are computed with the Stäckel-fudge formalism (Binney 2012; Sanders & Binney 2016). Point-estimate integrations span 4 Gyr with 2,001 stored trajectory samples, while the 5,000-realisation orbit Monte Carlo uses the same 4 Gyr interval with 1,001 stored samples per realisation. This interval samples many radial periods for the compact inner-reaching orbits ($T_r \lesssim 100\,\mathrm{Myr}$ at $R_{\mathrm{peri}} \sim 100\,\mathrm{pc}$) while keeping the orbit products descriptive rather than evolutionary. A static-potential re-integration of a random 100-star Tier A subset to 8 Gyr changes the sample medians by only +1.6 pc in R_{peri}, +0.010 kpc in R_{apo}, and +0.060 kpc in z_{max} relative to the 4 Gyr products. A rigidly rotating barred Hunter/Sormani variant (Sormani et al. 2022) is used as a sensitivity test for non-axisymmetric inner-Galaxy effects, not as the default catalogue definition.

In this manuscript, orbit-derived quantities such as eccentricity, R_{peri}, R_{apo}, z_{max}, energy, angular momentum, and actions are presented as outputs of the adopted model. They are informative, but they remain model-dependent, especially for stars whose inferred pericentres approach the central few tens of parsecs. Pericentres, apocentres, and maximum heights are measured directly from the integrated trajectories, not inferred from Stäckel-fudge actions. The reported actions are included for follow-up phase-space work; for the deepest plunging orbits their absolute values should be treated as approximate because Stäckel-fudge accuracy degrades in highly eccentric, strongly non-axisymmetric inner-Galaxy regimes (Wright & Binney 2025). This caveat is sharpened in Section 4.5 alongside the barred-potential sensitivity test.

2.5. *Matched-Control Sample*

To test whether orbit-derived properties vary continuously with Galactic rest-frame speed, we constructed a multi-band velocity-stratified control sample from the full Gaia DR3 6D catalogue. Four velocity bands were defined: 25–50, 50–100, 100–200, and 200–260 $\mathrm{km\,s^{-1}}$. Within each band, controls are kernel-weighted to match the slow catalogue's broad observability marginals in G, $\log d$, and $\sin|b|$, with the same valid-radial-velocity requirement as the slow-star sample. The effective control samples contain 1,955, 2,586, 2,788, and 2,749 stars for the four bands respectively.

We also tested a richer local reweighting of the same control-band catalogues using sky position (l, b), colour (BP–RP), RUWE, and parallax signal-to-noise in addition to the baseline observability marginals. This sensitivity test asks whether richer matching of the existing local control products changes the trend; it is not a fresh Gaia archive scan or a replacement control-parent construction. The monotonic increase in median pericentre with V_{grf} is preserved under the richer weights, so the main control result does not depend on the narrow choice of baseline matching variables.

The 25–50 $\mathrm{km\,s^{-1}}$ band yields fewer matched controls because the candidate pool at these velocities is intrinsically small: within the Gaia DR3 6D parent sample after the basic quality requirements used for control construction (valid radial velocity and parallax signal-to-noise > 5), there are only 4,763 stars in this range, compared with 39,853 at 50–100 $\mathrm{km\,s^{-1}}$ and over 1.4 million at 100–200 $\mathrm{km\,s^{-1}}$. The extreme sparsity of the low-velocity tail—7,622 stars below 50 $\mathrm{km\,s^{-1}}$ out of roughly 6.8 million stars in that quality-filtered 6D parent sample, or 0.11%—shows that this low-velocity tail is sparse within the observed Gaia DR3 6D catalogue. All controls were orbit-enriched using the same static Hunter+2024 potential and integration settings as the slow-star sample.

The stratified design tests how orbital properties vary with $V_{\rm grf}$, rather than comparing the slow-star sample to a single control window. The 25–50 $\mathrm{km\,s^{-1}}$ band is the adjacent kinematic comparison to the slow-star sample; the 100–200 $\mathrm{km\,s^{-1}}$ band samples a mixture of thick-disc and inner-halo kinematics; and the 200–260 $\mathrm{km\,s^{-1}}$ band brackets the circular speed at $R_\odot$ (232 $\mathrm{km\,s^{-1}}$) and therefore includes a substantial disc population.

2.6. *Catalogue Contents and Availability*

The accompanying catalogue includes Gaia `source_id`, the observables used for sample construction, adopted Galactocentric positions and velocities, $V_{\rm grf}$, quality flags, and the principal orbit-derived quantities reported here (eccentricity, $R_{\rm peri}$, $R_{\rm apo}$, $z_{\rm max}$, and related summary metrics). Orbit-derived fields are explicitly labelled as model-dependent outputs under the adopted potential rather than as direct Gaia observables.

3. EMPIRICAL RESULTS

3.1. *The Sample*

The final catalogue is tiered. The Gaia DR3 6D candidate pool contains 20,829 near-threshold sources retained from a broad full-parent scan; under the final distance and Solar-frame conventions, 2,591 have corrected point-estimate $V_{\rm grf} < 25\,\mathrm{km\,s^{-1}}$. The headline Tier A+B catalogue contains 517 stars with $P(V_{\rm grf} < 25\,\mathrm{km\,s^{-1}}) > 0.84$, while the broader Tier A+B+C catalogue contains 1,835 stars with $P > 0.50$. Unless otherwise stated, descriptive orbit-derived medians below are reported for Tier A+B+C and high-confidence observational-quality follow-up quantities for Tier A+B.

3.2. *The Headline Catalogue Survives Conservative Quality Cuts*

The Gold observational-quality subset starts from the Tier A+B headline catalogue and then applies only observational quality cuts. The catalogue Gold subset contains 397 stars after requiring RUWE < 1.4, acceptable RVS quality, parallax signal-to-noise > 10, and $\sigma_d/d < 0.15$. This gives a compact follow-up list whose definition does not depend on the orbit model.

Unresolved binaries and problematic single-epoch radial velocities can perturb Gaia RVS velocities on scales comparable to the 25 $\mathrm{km\,s^{-1}}$ threshold. We therefore treat RUWE and RVS-quality screens as catalogue diagnostics rather than proof of single-star centre-of-mass velocities. In the catalogue products, 27/517 Tier A+B stars ($5.2^{+1.1}_{-0.9}$%) and 55/1,835 Tier A+B+C stars (3.0 ± 0.4%) have RUWE > 1.4 (Jeffreys 68% intervals throughout); the Gold subset removes these systems, and the poor-RVS exclusion test in Section 4.5 shows that the compact radial-orbit summary is unchanged after removing flagged RVS solutions.

In this section we restrict the robustness check to empirical observables and simple descriptive properties of the sample. The corresponding effect of the Gold cut on the model-derived orbit summaries is reported separately in Section 4.5.

Table 5. "Gold observational-quality" subset: Tier A+B stars satisfying conservative observational quality cuts only — RUWE < 1.4, RVS quality flag = "ok" (Katz et al. 2023), parallax SNR > 10, $\sigma_d/d < 0.15$, and Tier A+B membership. Dynamical assumptions do not enter this subset; it is intended for observational follow-up that does not depend on the adopted potential.

Selection criterion	Count
Tier A+B (P> 0.84) base	517
& RUWE < 1.4	490
& RVS quality = "ok"	460
& parallax SNR > 10	397
& $\sigma_d/d < 0.15$	397
Final "gold observational-quality"	**397**
Median $V_{\rm grf}$ ($\mathrm{km\,s^{-1}}$)	15.7
Median G (mag)	13.6
Median distance (pc)	2,306

3.3. *Velocity-Threshold Membership Is Probabilistic*

The corrected point-estimate catalogue contains 2,591 stars below $V_{\rm grf} = 25\,\mathrm{km\,s^{-1}}$, but the threshold itself is not equally secure for every star. Monte Carlo propagation gives 276 stars in Tier A ($P > 0.95$), 517 stars cumulatively in Tier A+B ($P > 0.84$), and 1,835 stars cumulatively in Tier A+B+C ($P > 0.50$). The remaining 756 corrected point-estimate members form Tier D. The paper therefore uses Tier A+B as the headline catalogue and Tier A+B+C for broader population-level orbit summaries.

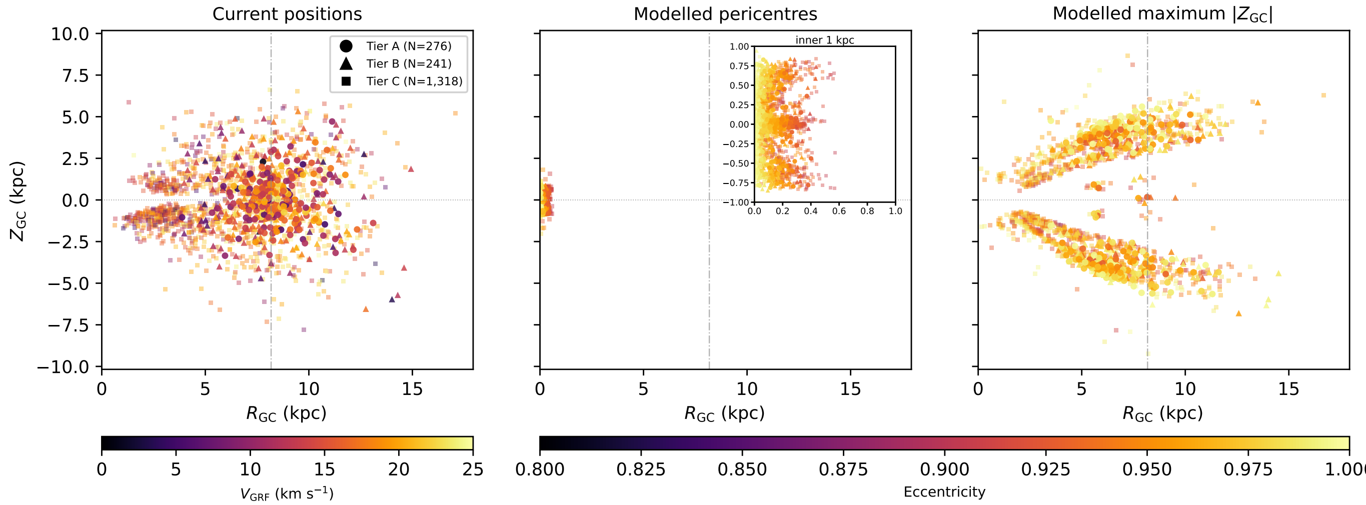

Figure 3. Edge-on orbital context for the Tier A+B+C slow catalogue ($N = 1{,}835$) in $R_{\rm GC}$ vs. $Z_{\rm GC}$. Current positions are observed coordinates, while the modelled pericentre and modelled maximum-$|Z_{\rm GC}|$ panels are model-derived locations in the adopted static Hunter+2024 potential. This broader Tier A+B+C sample is used for orbit-context visualisation; the Tier A+B headline catalogue is distinguished from the Tier C extension by marker shape. The panels show current positions, modelled pericentre positions, and modelled maximum-$|Z_{\rm GC}|$ points. Current positions are colour-coded by $V_{\rm grf}$; modelled pericentre and maximum-$|Z_{\rm GC}|$ positions are colour-coded by eccentricity, with the pericentre panel inset expanding the inner kiloparsec. The right panel shows the vertical envelope of the adopted model trajectories; apocentre radii are reported separately in Table 11 and Fig. 11. Marker shape encodes tier: circles (Tier A), triangles (Tier B), squares (Tier C). The vertical dot-dashed line marks $R_\odot = 8.178\,$kpc and the Galactic centre is at $R_{\rm GC} = 0$. Because $R_{\rm GC}$ is cylindrical radius, the pericentre panel and inset show central reach, not the signed side of the Galactic centre on which each passage occurs; that azimuthal side-of-centre information is retained in the face-on view of Fig. 4.

As a smooth-tail expectation check, we fit a trivariate Gaussian velocity ellipsoid to the four matched-control bands and normalise the resulting mock to the 10,078 control stars between 25 and $260\,\mathrm{km\,s^{-1}}$. This smooth DF predicts $N(< 25\,\mathrm{km\,s^{-1}}) = 76.5$ stars in the same control-normalised tail, whereas the probability-selected Tier A+B+C catalogue contains 1,835 stars. The implied excess factor is 24.0 in this diagnostic normalisation (Fig. 6). The number should not be read as a volume-complete population estimate; it is a local comparison against the observed Gaia DR3 matched-control velocity ellipsoid.

3.4. *Chemistry Does Not Define the Sample*

Metallicity information is used here only as a diagnostic of chemical heterogeneity and estimator bias, not for population assignment. The metallicity spread in the matched subset is broad enough that the catalogue should not be treated as a single chemical population.

These official Gaia DR3 GSP-Phot metallicities are low-resolution photometric estimates with documented systematics (Andrae et al. 2023; Recio-Blanco et al. 2023): in particular, GSP-Phot tends to overestimate metallicity for cool, metal-poor giants by 0.2–0.5 dex, which biases the present sample toward solar values. In the catalogue, GSP-Phot metallicities are finite for 1,214 Tier A+B+C stars and have a median [M/H] $= -0.25$. The spectroscopic alpha-subset discussed below is more metal-poor, with median [Fe/H] $= -0.80$. This offset is not interpreted as a population difference: it reflects both the known GSP-Phot metallicity systematics and the strongly non-random targeting of the APOGEE/GALAH overlap. The Zhang et al. (2023) XP catalogue remains the preferred larger photometric metallicity comparison for a future broader cross-match, but the present paper uses only the synchronized GSP-Phot and spectroscopic products as release materials.

3.5. *Chemical Properties from Spectroscopic Cross-Matching*

We cross-matched the full 20,829-source candidate pool against APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) using exact Gaia source identifiers rather than positional matching. The query returns 604 APOGEE matches and 489 GALAH matches across the candidate pool. Within the Tier A+B+C sample, 61 stars have

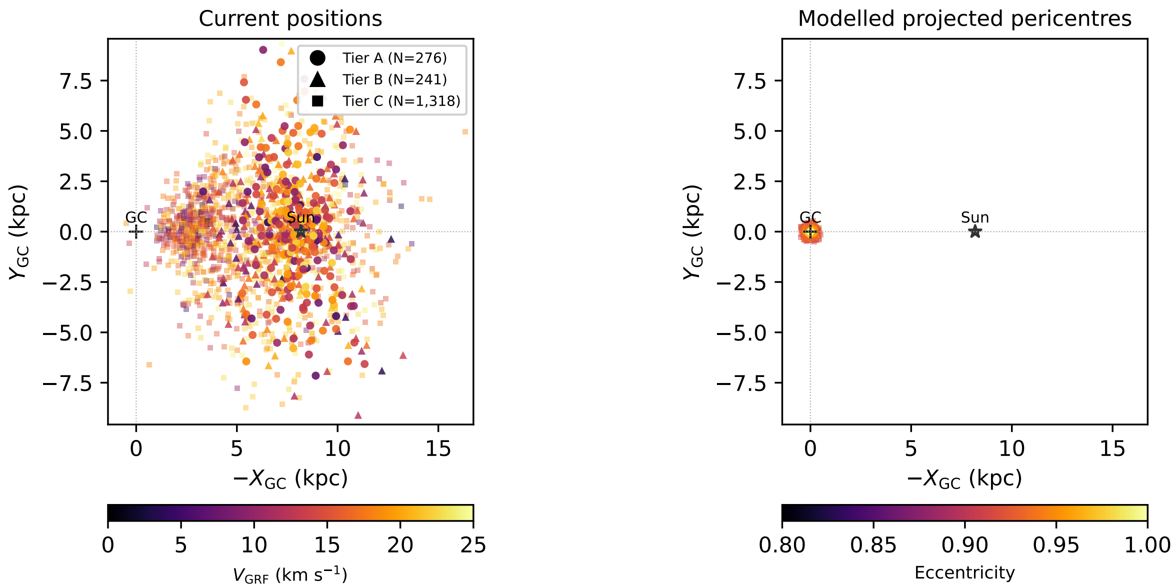

Figure 4. Face-on view of the Tier A+B+C slow catalogue ($N = 1{,}835$) in the displayed $-X_{\rm GC}$ vs. $Y_{\rm GC}$ plane, so that the Galactic centre is at left and the Sun lies to the right. As in Fig. 3, the broader Tier A+B+C orbit-summary sample is shown, with marker shape separating the Tier A+B headline catalogue from the Tier C extension. *Left:* current Galactocentric positions, colour-coded by $V_{\rm grf}$; the catalogue is observed in a Sun-centred local volume. *Right:* modelled, projected pericentre passage positions in the static Hunter+2024 potential, colour-coded by orbital eccentricity and shown on the same spatial scale as the left panel. Marker shape encodes tier (circle/triangle/square = A/B/C). The shared scale makes the inner-Galaxy reach of the catalogue directly visible: all point-estimate pericentres fall inside $R_{\rm GC} < 1\,$kpc in this fine trajectory sampling, with a median Tier A+B+C pericentre of 112 pc in the 40,001-point interpolated readout.

Table 6. Monte Carlo threshold-membership probabilities for the $V_{\rm grf} < 25$ km s^{-1} cut. Astrometric realisations use the Gaia DR3 parallax–proper-motion covariance, with sky position held fixed, plus an independent radial-velocity error, with distance drawn from the asymmetric Bailer-Jones+2021 photogeometric posterior. The candidate pool contains 20,829 Gaia DR3 sources; after the final correction and distance convention, 2,591 have corrected point-estimate $V_{\rm grf} < 25$ km s^{-1}. "Sources newly admitted beyond the legacy preselection" is the count of Tier A+B+C members that do not appear in the pre-v15 slow-velocity preselection list maintained during catalogue development; they are the additions enabled by the zero-point-corrected, propagated, probability-tiered pipeline reported here.

Diagnostic	Value
Candidate pool propagated through MC	20,829
Corrected point-estimate $V_{\rm grf} < 25$ km s^{-1}	2,591
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.95$ (Tier A)	276
$0.84 < P(V_{\rm grf} < 25$ km s$^{-1}) \leq 0.95$ (Tier B increment)	241
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.84$ (Tier A+B; headline)	517
$0.50 < P(V_{\rm grf} < 25$ km s$^{-1}) \leq 0.84$ (Tier C increment)	1,318
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.50$ (Tier A+B+C)	1,835
Tier D (point-estimate < 25 km s^{-1}, $P \leq 0.50$)	756
Median $P(V_{\rm grf} < 25$ km s$^{-1})$ for Tier A+B+C	0.71
Sources newly admitted beyond the legacy preselection (Tier A+B+C)	1,202

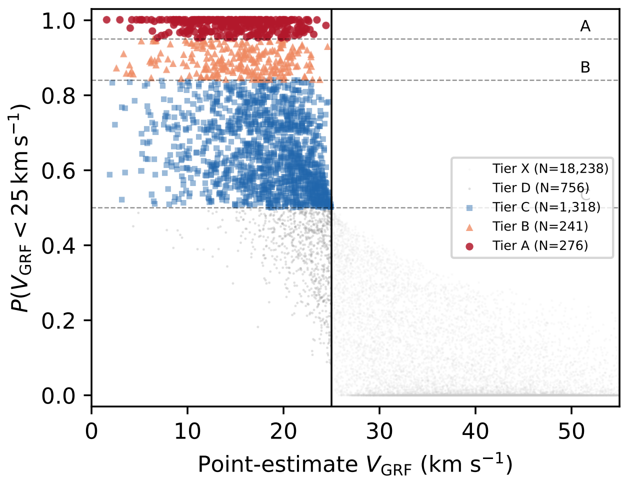

Figure 5. Velocity-threshold membership probability versus corrected point-estimate $V_{\rm grf}$ for the propagated Gaia DR3 candidate pool. The horizontal dashed lines mark the Tier A, Tier A+B, and Tier A+B+C probability boundaries; the vertical line marks the $25\,{\rm km\,s^{-1}}$ point-estimate selection boundary. The catalogue is defined by threshold probability, not by a hard point estimate alone.

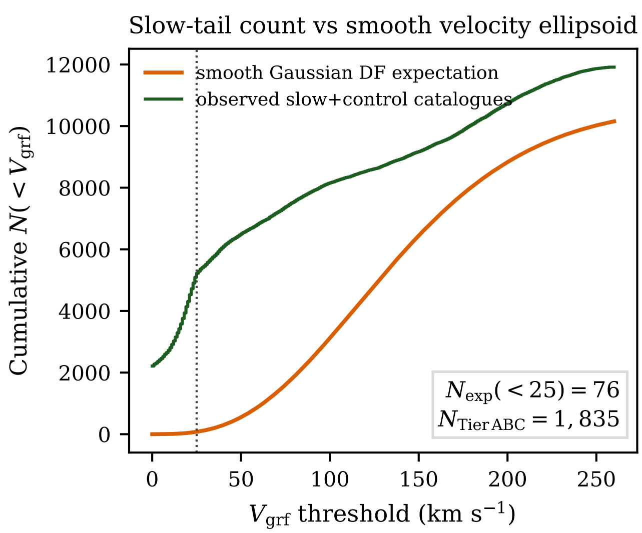

Figure 6. Observed cumulative slow-tail count compared with a smooth trivariate-Gaussian velocity ellipsoid fit to the matched-control catalogues and normalised over 25–$260\,{\rm km\,s^{-1}}$. The excess below the adopted $25\,{\rm km\,s^{-1}}$ selection threshold motivates treating the low-$V_{\rm grf}$ catalogue as a statistically distinct low-velocity tail of the observed Gaia DR3 6D distribution, not as evidence for a physical discontinuity at exactly $25\,{\rm km\,s^{-1}}$. Absolute pipeline normalisation is provided separately by the controlled injection-recovery test of Section 4.7.

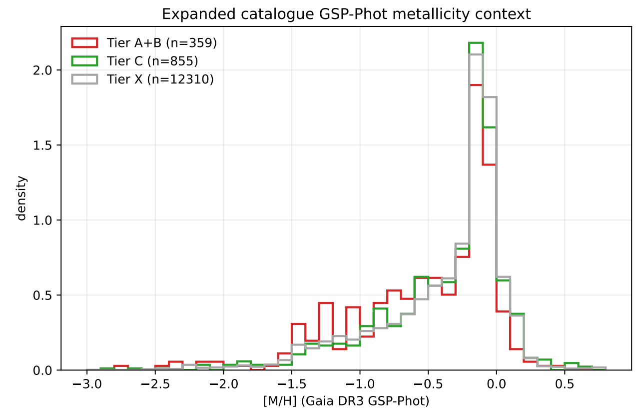

Figure 7. Gaia GSP-Phot metallicity distribution for the catalogue, shown as a low-resolution contextual diagnostic. These values are not used for population-region assignment.

usable APOGEE α-element abundances, 55 have usable GALAH abundances, and after survey-priority deduplication 113 unique Tier A+B+C stars carry the joint [Fe/H] + α-element abundance combination required for the literature-region comparisons reported in Table 7. This 113-star α-classified subset is dominated by luminous giants in the APOGEE and GALAH footprints; it is therefore a selective but useful chemical window, not a chemically complete census.

Within this alpha-classified subset, 53 stars fall in a Splash-like region, 23 in a Gaia-Sausage/Enceladus-like region, 19 in an Aurora-like region, 2 in a disc-like contaminant region, and 16 are not assigned to these literature regions. These fractions are reported only within the 113-star alpha-classified subset. They are not used as whole-catalogue population fractions, because the remaining Tier A+B+C stars lack the high-resolution abundance information required for that assignment in the current release products.

The broader metallicity context is still informative. Gaia GSP-Phot metallicities are available for many candidates, but the comparison in Section 3.4 shows that GSP-Phot is biased toward solar metallicity for the cool, metal-poor giants that dominate this sample. We therefore use GSP-Phot only as a low-resolution heterogeneity check and rely on spectroscopic abundances for subset-qualified population language. Photometric metallicity alone does not supply the alpha-element information needed for the Splash/GSE/Aurora comparison cuts.

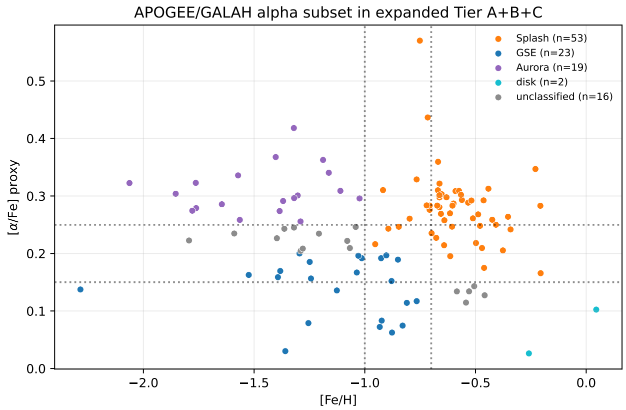

Figure 8. [Mg/Fe] (APOGEE) and [α/Fe] (GALAH) vs. [Fe/H] for the α-classified spectroscopic subset within Tier A+B+C ($N = 113$). Literature-region labels are shown only as contextual comparisons and are not treated as whole-catalogue population assignments.

Table 7. High-resolution spectroscopic chemistry available for the Tier A+B+C catalogue. APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) provide the subset used for α-based literature-region comparisons. Matches come from a full exact-source-id VizieR TAP cross-match against the 20,829-source candidate pool. Gaia GSP-Phot metallicities are discussed only as a biased low-resolution contextual estimator in the text, not as population assignments.

Source	N	Median [Fe/H]
APOGEE DR17, usable α	61	−0.80
GALAH DR3, usable α	55	−0.83
unique stars after deduplication	113	−0.80
Splash classified	53	−0.62
GSE classified	23	−1.03
Aurora classified	19	−1.38
disc-like contaminant	2	−0.11
unclassified in these regions	16	−1.14

Table 8. Subset-qualified chemodynamic context for the Tier A+B+C catalogue. Only stars with both [Fe/H] and [α/Fe] from APOGEE or GALAH ($n = 113$) are placed in literature comparison regions following Belokurov et al. (2020), Helmi et al. (2018), and Belokurov & Kravtsov (2022). The remaining 1,722 stars are not assigned to a population in this catalogue paper. Fraction errors are Beta($k + \frac{1}{2}, n - k + \frac{1}{2}$) Jeffreys 68% intervals on the 113-star subset (Cameron 2011); with $n = 113$ they are large, which is itself an argument against extrapolating to the full catalogue.

Population	N	Fraction	[Fe/H]
Splash	53	$46.9^{+4.7}_{-4.6}$%	−0.62
GSE	23	$20.4^{+4.0}_{-3.5}$%	−1.03
Aurora	19	$16.8^{+3.8}_{-3.2}$%	−1.38
Disc-like	2	$1.8^{+1.7}_{-0.9}$%	−0.11
Unclassified	16	$14.2^{+3.6}_{-2.9}$%	−1.14
α-classified subtotal	113	100.0%	—
not α-classified	1,722	—	—

3.6. *Sky Distribution, Observer Geometry, and Selection*

The sample is not isotropically distributed on the sky. In Galactic coordinates, 39.0 ± 1.1% of the Tier A+B+C stars lie toward the Galactic-centre direction ($l \in [330°, 30°]$), while only $6.2^{+0.6}_{-0.5}$% lie toward the anti-centre direction ($l \in [150°, 210°]$). Latitude coverage is broad, from about −85° to +87°, with a slight southern bias in the median latitude.

This concentration is one of the strongest descriptive features of the observed catalogue. It likely reflects some combination of heliocentric observing geometry, the Gaia radial-velocity selection function, extinction and crowding effects, and the radial character of the underlying trajectories. The velocity-stratified controls are also centre-weighted, indicating that the sky distribution is partly a generic property of observed Gaia 6D stars at these distances.

The Castro-Ginard et al. (2023) Gaia DR3 RVS selection function (extending the EDR3 RVS sample-completeness framework of Everall & Boubert 2022) is therefore used as a limitation and contextual observability diagnostic for the observed catalogue rather than as a precise deprojection of the underlying Milky Way population. We evaluate the catalogue with the GaiaUnlimited DR3-RVS implementation and release the per-star selection-function table, but inverse-selection weighted counts are not used as headline population counts.

Table 9. Gaia DR3 RVS selection-function diagnostics for the catalogue. The catalogue is selected from the observed Gaia DR3 6D/RVS subset and is therefore not a volume-complete Milky Way census. $N_{\rm valid}$ is the number of sources for which the GaiaUnlimited DR3-RVS implementation (Castro-Ginard et al. 2023) returned a finite selection-function weight at the source's $(G, BP{-}RP, l, b)$ cell; the $N - N_{\rm valid}$ deficit is dominated by cells with zero parent count. $f_{n<10}$ is the fraction of $N_{\rm valid}$ sources whose underlying GaiaUnlimited cell carries a parent count below 10 (the regime in which the Beta$(k{+}1, n{-}k{+}1)$ posterior remains competitive with its Beta$(1,1)$ prior: posterior standard error $\gtrsim 0.16$ for $n < 10$); $f_{\sum,n<10}$ is the share of $\sum 1/S_{\rm RVS}$ contributed by those prior-dominated cells. The lower block restricts each tier to the inner-Galaxy quadrant $l \in [330°, 30°]$ where the catalogue concentrates (Section 3.6). Inverse-selection weighted sums are reported only as contextual observability diagnostics and are not used as headline population counts.

Sample	N	$N_{\rm val}$	$\Sigma 1/S$	$\tilde{S}$	$f_{<10}$	$f_{\Sigma,<10}$
Full sample						
Candidate pool	20,829	19,999	37,256.3	0.50	–	–
Tier A	276	274	435.7	0.67	80.3%	86.7%
Tier B	241	237	408.4	0.50	87.8%	91.6%
Tier C	1,318	1,261	2,324.0	0.50	86.3%	89.5%
Tier A+B	517	511	844.0	0.67	83.8%	89.0%
Tier A+B+C	1,835	1,772	3,168.1	0.50	85.6%	89.4%
Inner-Galaxy quadrant, $l \in [330°, 30°]$						
Tier A (inner)	35	34	47.0	0.67	58.8%	66.2%
Tier A+B (inner)	79	78	122.6	0.67	66.7%	73.8%
Tier A+B+C (inner)	715	664	1,260.1	0.50	82.7%	85.2%

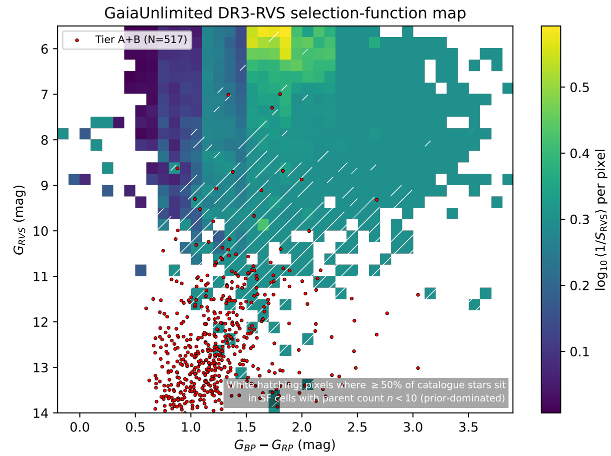

Figure 9. GaiaUnlimited DR3-RVS (Castro-Ginard et al. 2023) selection-function weight map for the expanded candidate pool, queried in the same $(G_{RVS}, BP{-}RP, l, b)$ convention as the pipeline. Cell colour is the per-pixel mean $\log_{10}\langle 1/S_{\rm RVS}\rangle$; red points mark the Tier A+B headline catalogue ($N = 517$). White hatching flags pixels in which $\geq$ 50% of catalogue stars sit in GaiaUnlimited cells with parent count $n < 10$, the regime where the Beta$(k{+}1, n{-}k{+}1)$ posterior is prior-dominated. The Tier A+B locus overlaps the hatched region substantially, consistent with the $f_{n<10} =$ 83.8% of stars and 89.0% of $\sum 1/S_{\rm RVS}$ reported in Table 9.

4. ORBIT-DERIVED PROPERTIES IN THE ADOPTED POTENTIAL

Within the adopted static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits, and this result is robust across a suite of diagnostic tests. The following subsections present the primary orbit-derived properties, compare them against velocity-stratified controls, describe how the sample's properties vary with orbital scale, and assess robustness to measurement uncertainty and potential choice.

Table 10 identifies the sampling scheme behind the repeated pericentre and central-reach statistics used below. For the catalogue, the primary orbit summary uses 5,000-realisation orbit Monte Carlo propagation for every Tier A+B+C star in the adopted static Hunter+2024 potential. The velocity-stratified control comparison, halo-mass variants, barred potential variants, and central-reach counts are retained as explicitly labelled point-estimate diagnostics.

4.1. *In the Adopted Model, the Sample Is Strongly Radially Biased*

Within the adopted static Hunter+2024 potential, the orbit-derived picture is strongly radial. For the Tier A+B+C catalogue (N = 1,835), the median of the per-star posterior-median eccentricities is 0.949. All 1,835 stars have $e > 0.8$, 49.2 ± 1.2% have a posterior-median $e > 0.95$, and $0.9{\pm}0.2$% have a posterior-median $e > 0.99$ from the 5,000-realisation Monte Carlo orbit product (Table 11; Jeffreys 68% intervals).

The corresponding fractions obtained from a single deterministic point-estimate orbit integration in the same potential (Table 18) are higher: 65.7 ± 1.1% of Tier A+B+C stars have $e > 0.95$ under point-estimate integration. The difference between the Monte Carlo posterior-median fraction (49.2%) and the point-estimate fraction (65.7%) is the expected consequence of convolving the measurement-error distribution into a quantity that is bounded above by unity: the per-star posterior medians regress slightly toward lower e relative to the noise-free point estimate. Both numbers describe the same catalogue; we report the posterior-median fraction as the headline value because it propagates the observational uncertainties explicitly.

The same Monte Carlo orbit product gives a Tier A+B+C median $R_{\rm peri}$ of 151 pc and a median $R_{\rm apo}$ of 7.45 kpc. The maximum posterior median $R_{\rm apo}$ in the present Tier A+B+C integrations is 17.85 kpc.

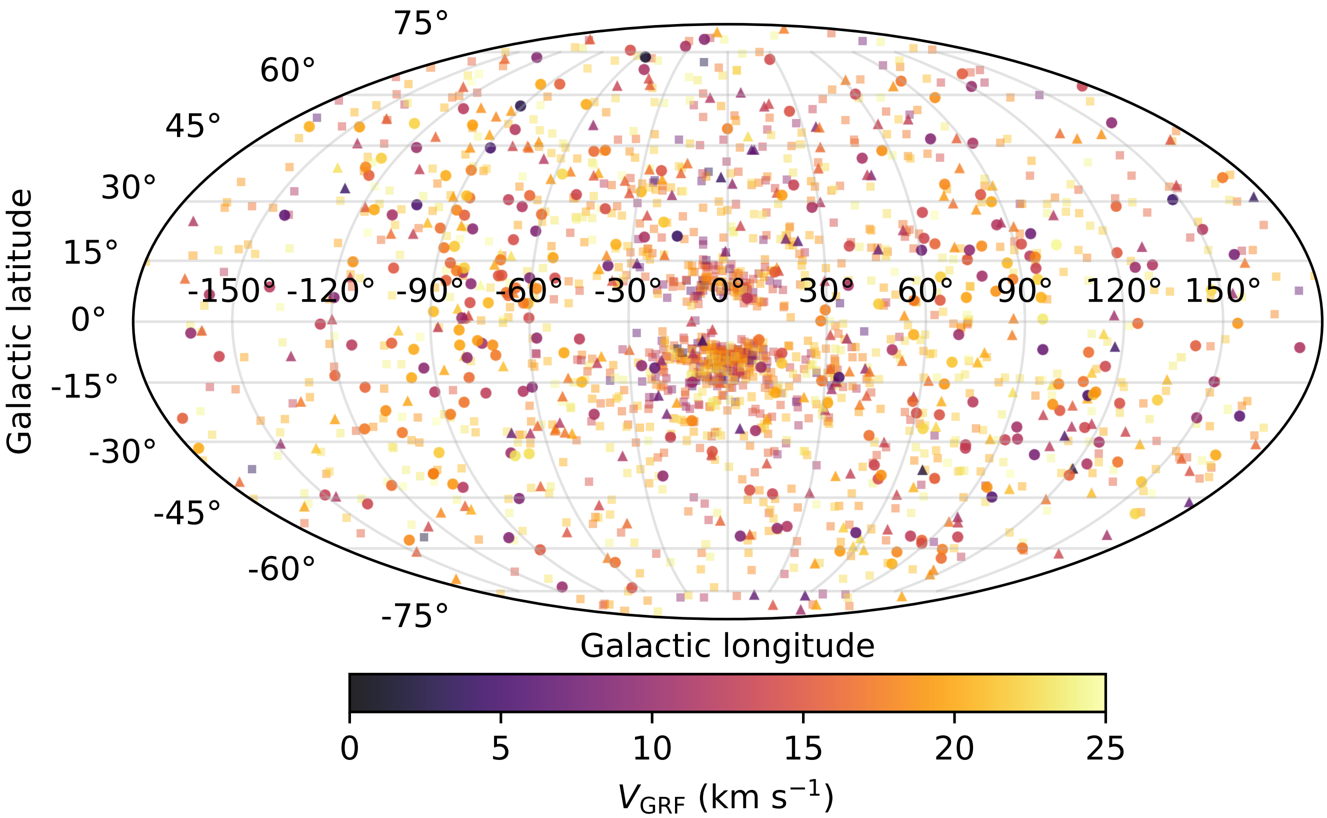

Figure 10. Aitoff projection of the Tier A+B+C slow catalogue ($N = 1{,}835$) in Galactic coordinates ($l = 0$ centred, l increasing to the left), colour-coded by V_{grf}. Marker shape encodes tier (circle/triangle/square = A/B/C). The catalogue concentrates strongly toward the inner Galaxy.

Table 10. Ledger for repeated pericentre and central-reach statistics. Each row refers to a different sampling scheme or sensitivity test, so the numerical values are not interchangeable.

Use in paper	Sampling scheme	Quantity reported	Interpretation
Primary orbit summary	5,000-realisation orbit Monte Carlo in the static Hunter+2024 potential	Median $R_{\mathrm{peri}} = 151$ pc, median $R_{\mathrm{apo}} = 7.45$ kpc, median $e = 0.949$	Uncertainty-propagated catalogue summary used for Table 11.
Control comparison	Slow-catalogue point estimate; existing controls re-weighted to catalogue observability marginals	Median R_{peri} rises from 112 pc (slow) to 2.26 kpc (200–260 km s^{-1})	Supports the endpoint interpretation in Table 12.
Central reach	Point-estimate static and barred integrations	Static: 824 stars with $R_{\mathrm{peri}} < 100$ pc; barred: 1,543	Shows that the catalogue remains strongly inner-reaching.
Halo/bar/distance sensitivity	Point-estimate halo and bar-speed reruns; BJ/parallax coupling stress test	Halo variants: median $R_{\mathrm{peri}} = 109$–116 pc; bar variants: 6–24 bridgers	Shows the main radial-orbit result is stable while inner-Galaxy bridgers remain potential-sensitive.

Table 11. Summary of orbit-derived properties for the Tier A+B+C catalogue (N = 1,835) under 5,000-realisation Monte Carlo orbit propagation in the Hunter+2024 axisymmetric Galactic potential. Reported values are summaries of the per-star posterior medians; $R_{\rm peri}$, $R_{\rm apo}$, and $z_{\rm max}$ are computed in the cylindrical galactocentric frame over a 4 Gyr forward integration. Eccentricity-fraction uncertainties are Beta($k+\frac{1}{2}, n-k+\frac{1}{2}$) Jeffreys 68% intervals (Cameron 2011); the corresponding point-estimate fractions are in Table 18.

Quantity	Value
Median posterior-median e	0.949
Posterior-median $e > 0.8$ (%)	100.0
Posterior-median $e > 0.95$ (%)	49.2 ± 1.2
Posterior-median $e > 0.99$ (%)	0.9 ± 0.2
Median $R_{\rm peri}$ (pc)	151
$R_{\rm peri}$ p16 (pc)	89
$R_{\rm peri}$ p84 (pc)	274
Median $R_{\rm apo}$ (kpc)	7.45
Maximum $R_{\rm apo}$ (kpc)	17.85
Median $z_{\rm max}$ (kpc)	3.38

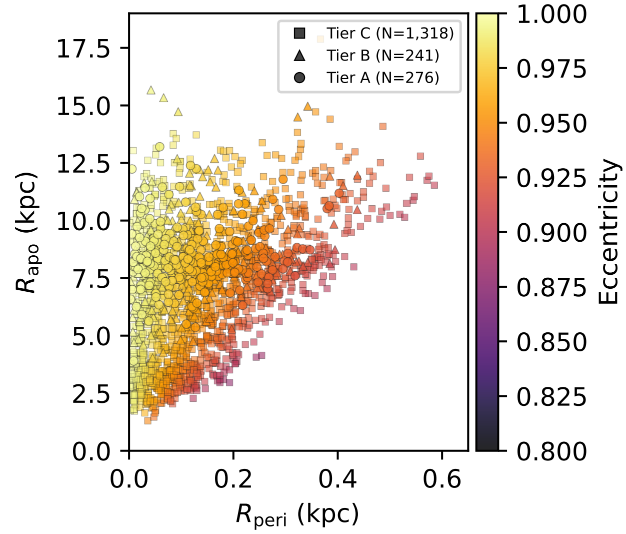

Figure 11. Pericentre vs. apocentre for the Tier A+B+C slow catalogue (N = 1,835), colour-coded by eccentricity, in the static Hunter+2024 potential. Axis ranges are set to where the adopted-model point-estimate integrations place the data: all 1,835 stars have $R_{\rm peri}$ < 1 kpc and nearly all have $R_{\rm apo}$ < 18 kpc. The bridger sensitivity test (§4.4) is reported in the text and in Table 15, not on this panel.

These values confirm that the low-speed catalogue is not dominated by ordinary near-circular disc kinematics.

These quantities should be read as outputs of the adopted model rather than as direct Gaia observables. They are useful because they provide a coherent dynamical interpretation of the slow state. Pericentre estimates below roughly 200 pc should be treated as indicative, because the real inner Galaxy is barred and non-axisymmetric on those scales; the barred Hunter/Sormani variant in Section 4.5 brackets this sensitivity.

4.2. *Matched-Control Comparison*

A velocity-stratified, observability-matched control comparison provides the clearest test of whether the slow-star orbit result merely reflects the typical orbital structure of Gaia stars at similar distances, or whether it varies systematically with Galactic rest-frame speed. The controls are kernel-weighted to match the slow catalogue's broad observability marginals. In the catalogue, the slow-sample point-estimate median $R_{\rm peri}$ is 112 pc. The re-weighted control medians rise from 140 pc in the adjacent 25–50 km s^{-1} band to 428 pc, 1.40 kpc, and 2.26 kpc in the three faster bands.

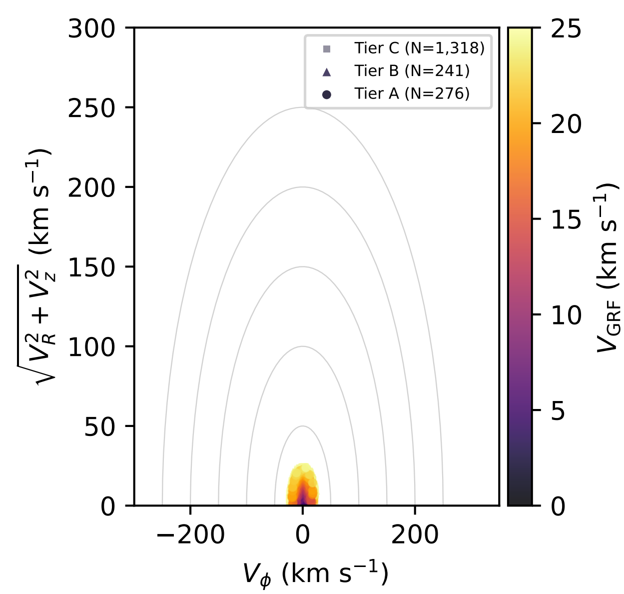

Figure 12. Toomre diagram (V_ϕ vs. $\sqrt{V_R^2 + V_z^2}$) for the Tier A+B+C slow catalogue (N = 1,835), colour-coded by $V_{\rm grf}$. The sample is concentrated at low azimuthal velocity, far from the disc locus.

This preserves the endpoint interpretation: the slow catalogue sits at the compact-pericentre end of a continuous velocity-stratified sequence rather than at a sharp physical boundary.

The existing local control catalogues were also reweighted more aggressively in l, b, BP–RP, RUWE, and parallax signal-to-noise, in addition to the baseline $(G, \log d, \sin|b|)$ matching. The same qualitative pericentre trend is preserved, so the control result is not driven by the narrow choice of baseline matching variables.

The same comparison provides a direct check on bar-resonance accumulation. The slow sample is not uniquely enhanced at the OLR: its weighted OLR fraction is 1.85%, lower than the 100–200 and 200–260 $\mathrm{km\,s^{-1}}$ controls and only modestly higher than the adjacent low-velocity controls. Barred integrations therefore remain important for individual inner-Galaxy orbits and for the bridger sensitivity test, but the matched-control comparison does not support OLR accumulation as the primary explanation for the catalogue. Figure 19 in Section 4.6 places the catalogue in L_z–J_R action space alongside the inner Lindblad, corotation, and outer Lindblad loci for the full Ω_p grid; the Tier A+B+C distribution concentrates near $L_z \approx 0$ rather than at any single pattern-speed-resonance pile-up.

The full eccentricity CDFs reinforce the same point. A two-sample Anderson–Darling comparison (Scholz & Stephens 1987) between the Tier A+B+C static-potential eccentricities and the adjacent 25–50 $\mathrm{km\,s^{-1}}$ matched-control CDF gives $A^2 = 192.27$ with asymptotic $p < 0.001$. The comparison to the smooth-DF mock associated with Fig. 6 gives $A^2 = 36.97$, again with asymptotic $p < 0.001$. Thus the near-unity eccentricity concentration is not reproduced by either the adjacent observed control band or the control-normalised smooth velocity ellipsoid.

Table 12. Matched-control comparison across $V_{\rm grf}$ velocity bands. Each band is integrated in the Hunter+2024 axisymmetric potential; the kernel-weighted control samples are re-weighted to match the slow catalogue's $(G, \log d, \sin|b|)$ marginals. Bar-resonance fractions are evaluated at $\Omega_R/(\Omega_\phi - \Omega_p) = 2$ (OLR) and 4 (4:1 ultraharmonic) within ± 0.3.

Sample	N	$R_{\rm peri}$	$R_{\rm apo}$	OLR	4:1
slow A+B+C	1,835	112	7.4	1.85	1.80
25–50	1,955	140	3.2	1.09	0.29
50–100	2,586	428	3.7	0.51	0.57
100–200	2,788	1,397	4.0	2.55	0.96
200–260	2,749	2,260	4.8	2.93	1.22

Control-row labels are $V_{\rm grf}$ bands in $\mathrm{km\,s^{-1}}$; $R_{\rm peri}$ is in pc and $R_{\rm apo}$ is in kpc. The slow row uses the parent-complete Tier A+B+C catalogue. Control rows use the same integrated control catalogues re-weighted against the slow-sample observability marginals.

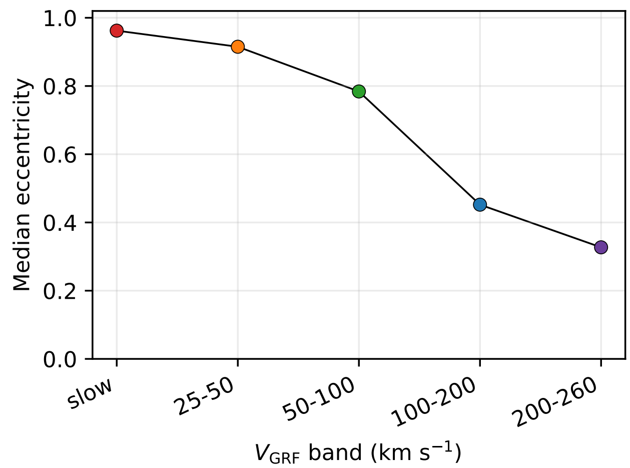

Figure 13. Median eccentricity as a function of Galactic rest-frame velocity for the Tier A+B+C slow catalogue ($N =$ 1,835) and the four observability-matched control bands. Points show the band medians and 16th–84th percentile ranges after the control bands are kernel-weighted against the slow catalogue. The monotonic decline from $e \sim 1.0$ at low $V_{\rm grf}$ to moderate eccentricity at 200–260 $\mathrm{km\,s^{-1}}$ matches the band-by-band medians in Table 12.

4.3. *Dwell-Time Analysis in the Adopted Potential*

A dwell-time calculation in the adopted static Hunter+2024 potential helps quantify what the low-$V_{\rm grf}$ selection captures. Tier A stars spend a median 28.3% of their integrated orbital time inside $R < 2\,\mathrm{kpc}$, while Tier C stars spend 19.7% there. Conversely, the median time fraction in the solar annulus ($7 < R < 9\,\mathrm{kpc}$) rises from 11.6% in Tier A to 16.2% in Tier C. The high-confidence slow stars are therefore not merely solar-radius apocentre snapshots; they are strongly inner-reaching orbits observed during a low-speed phase.

Table 13. Tier-resolved orbital dwell statistics in the Hunter+2024 potential. Reported are the per-tier medians of the time fraction spent within selected cylindrical radius shells over a 4 Gyr integration, indicating that high-confidence slow-$V_{\rm grf}$ stars spend a disproportionate fraction of their trajectory inside $R < 2\,\mathrm{kpc}$ despite being observed at the solar circle.

Shell	Tier A	Tier B	Tier C
Median $f(R < 1\,\mathrm{kpc})$ (%)	14.2	11.8	9.4
Median $f(R < 2\,\mathrm{kpc})$ (%)	28.3	24.1	19.7
Median $f(R < 5\,\mathrm{kpc})$ (%)	65.4	60.8	55.0
Median $f(7 < R < 9\,\mathrm{kpc})$ (%)	11.6	13.5	16.2

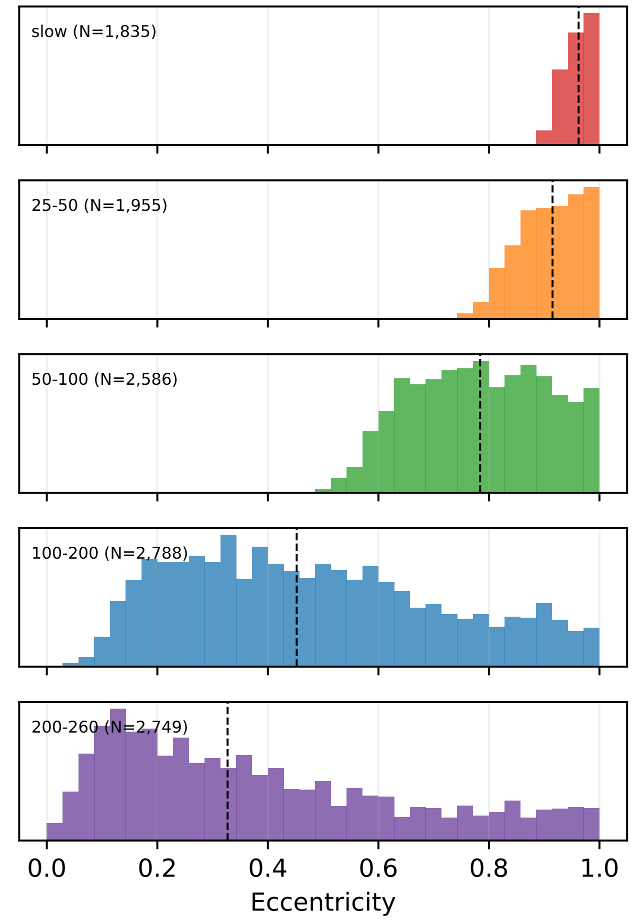

Figure 14. Eccentricity histograms for each of the five GRF velocity bands, stacked vertically. The slow Tier A+B+C band ($N = 1{,}835$) shows the extreme near-unity peak; the eccentricity distribution broadens monotonically through the four control bands toward 200–260 $\mathrm{km\,s^{-1}}$.

4.4. *Inner-Galaxy Reach and Selection Limits*

Although the catalogue shares the defining low-V_{grf} state, the stars span a range of model-derived orbital scales. The Tier A+B+C catalogue has a median static-potential pericentre of 151 pc in the orbit Monte Carlo posterior medians, and all 1,835 posterior-median pericentres remain inside 1 kpc. In the dedicated point-estimate central-reach diagnostic, R_{peri} is the cylindrical in-plane pericentre, $\min\sqrt{x^2+y^2}$, i.e. closest approach to the Galactic spin axis rather than to Sgr A* as a point. With this definition, 824 stars have static $R_{\mathrm{peri}} < 100\,\mathrm{pc}$ and all 1,835 have static $R_{\mathrm{peri}} < 1\,\mathrm{kpc}$. This fact is visible in the right panel of Fig. 4: every star's model pericentre passage falls within the inner kpc of the Galaxy, despite the catalogue being observed in a Sun-centred local volume (left panel). In the barred sensitivity integration the corresponding

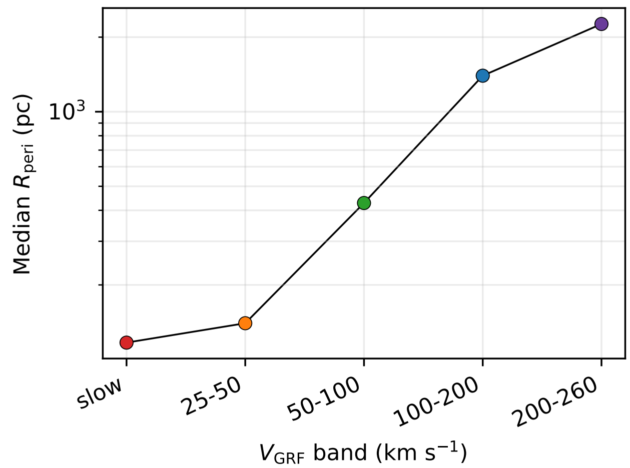

Figure 15. Median pericentre as a function of Galactic rest-frame velocity for the slow Tier A+B+C catalogue ($N = 1{,}835$) and the four observability-matched control bands. The slow band sits at the small-R_{peri} end of a smooth kinematic sequence; per-band pericentre medians are reported in Table 12.

cylindrical point-estimate count is 1,543 stars with $R_{\mathrm{peri}} < 100\,\mathrm{pc}$; those barred quantities are used as sensitivity diagnostics rather than as the default catalogue values.

The most extreme central-passage claims require additional caution. In the point-estimate products, the closest static and barred trajectories reach within 3.3 and 5.0 pc of the Galactic centre, respectively. Dedicated 5,000-realisation Monte Carlo propagation of the closest static and barred seeds shows that no candidate has $P(r_{\mathrm{sph,min}} < 10\,\mathrm{pc}) > 0.5$, while six candidates have $P(r_{\mathrm{sph,min}} < 100\,\mathrm{pc}) > 0.5$. These probabilities use the spherical Galactocentric closest approach, $r_{\mathrm{sph}} = \sqrt{x^2+y^2+z^2}$, and are Monte Carlo posterior probability counts, not the point-estimate cylindrical R_{peri} counts above. Since $r_{\mathrm{sph}} \geq \sqrt{x^2+y^2}$, the Sgr A* set is a stricter subset of the in-plane central-reach set. We therefore describe these as candidate inner-Galaxy approachers, not as a confirmed population of Sgr A*-passing stars. A three-point quadratic interpolation of $r^2_{\mathrm{sph}}(t)$ around each sampled trajectory minimum is included in the probabilities in Table 16. Figure 16 shows the corresponding cumulative probability curves for the 20 closest point-estimate seeds in the static and barred-default integrations.

The closest-approach calculation follows the same general philosophy as Gaia DR3 stellar-encounter work, where orbit integration and uncertainty propagation are used to identify robust close-passage candidates rather than to treat nominal closest approaches as definitive (Bailer-Jones 2022).

As an external source-id check, none of the 517 Tier A+B stars overlaps the 61 Gaia DR3 close-encounter entries in Bailer-Jones (2022); Table 14 reports the null cross-match explicitly.

The radial-reach tail is also potential-sensitive. Three Tier A+B+C stars (Poisson 68% band 1.4–5.9) satisfy the bridger criterion ($R_{\rm peri} < 2\,{\rm kpc}$ and $R_{\rm apo} > 15\,{\rm kpc}$) in the static Hunter+2024 integration. The barred default integration produces 22 such bridgers (68% Poisson band 17.4–27.7); across the pattern-speed grid explored in Section 4.5, the count ranges from 6 to 24. Because the absolute count varies by a factor of several between the static and default-barred integrations and remains Ω_p-sensitive within the barred family, we treat bridgers as bar-sensitive orbital-geometry candidates whose count is intrinsically potential-dependent, rather than as a robust population count or a merger-debris census.

Table 14. Overlap with the Bailer-Jones (2022) Gaia DR3 close-encounter catalogue.

Catalogue subset	BJ22 matches
Tier A+B	0

Note. BJ22 values are the median perihelion distances from VizieR catalogue J/ApJ/935/L9, Table 12. Distances in pc. The comparison uses Gaia DR3 `source_id`; this paper's values are present-day point-estimate orbit minima in the default static and barred diagnostic potentials.

Table 15. Inner-galaxy reach of the Tier A+B+C catalogue (N = 1,835). Counts are 40,001-point, parabola-interpolated point-estimate diagnostics for stars with small cylindrical in-plane pericentre $R_{\rm peri} = \min\sqrt{x^2+y^2}$, very small minimum spherical Galactocentric distance $r_{\rm sph}$, and "extreme reach" trajectories spanning the disc-to-halo radial range; the Monte Carlo orbit summary is reported separately in Table 11. The Static and Barred columns integrate under the Hunter+2024 static and barred potentials respectively; the Portail+2017 column uses the made-to-measure inner-Galaxy barred potential (Sormani-Vasiliev Agama recipe) rotated at $\Omega_p = 37.5\,{\rm km\ s^{-1}\ kpc^{-1}}$ to bracket the bridger count against an independent gravitational background. Bridger-row uncertainties are Gehrels (1986) Poisson 68% intervals. The barred median pericentre readout remains sampling-sensitive at tens-of-pc scales, so threshold counts are the more stable barred inner-reach diagnostic.

Diagnostic	Static	Hunter+24	Portail+17
$R_{\rm peri} < 100\,{\rm pc}$	824	1,543	1,571
$R_{\rm peri} < 50\,{\rm pc}$	453	1,431	1,469
$R_{\rm peri} < 10\,{\rm pc}$	88	629	607
Min $r_{\rm sph} < 100\,{\rm pc}$	252	760	721
Min $r_{\rm sph} < 10\,{\rm pc}$	4	7	3
Bridgers	$3^{+2.9}_{-1.6}$	$22^{+5.7}_{-4.6}$	$13^{+4.7}_{-3.6}$

Table 16. Candidate Sgr A* approachers from the Tier A+B+C catalogue. Candidate rows are seeded from the closest point-estimate static and barred integrations, then re-evaluated with 5,000-realisation Monte Carlo orbit propagation and three-point quadratic interpolation of $r^2_{\rm sph}(t)$ around each sampled minimum.

Gaia DR3 source_id	Tier	Potential	$r_{\rm sph,min}$ p16 (pc)	p50	p84	$P(<10\,{\rm pc})$ / $P(<100\,{\rm pc})$
2071829452678601472	A	static	12	24	46	0.107 / **0.999**
4112655451932785152	C	barred	14	31	65	0.071 / **0.923**
5602125925243507840	B	barred	32	68	135	0.014 / **0.689**
6743739021463974912	C	barred	30	71	154	0.017 / **0.661**
3673477423666158080	B	barred	29	70	252	0.027 / **0.639**
4656611002687082496	C	static	24	68	193	0.048 / **0.639**
5884580841726690688	C	barred	33	102	159	0.012 / 0.492
4125438271037670272	C	static	21	152	275	0.071 / 0.359
197752599391374208	B	static	49	180	389	0.014 / 0.280
4109059847125694848	C	static	83	290	411	0.012 / 0.209

NOTE—No catalogue candidate satisfies $P(r_{\rm sph,min} < 10\,{\rm pc}) > 0.5$. Six candidates satisfy the softer $P(r_{\rm sph,min} < 100\,{\rm pc}) > 0.5$ criterion. These are spherical Galactocentric closest-approach probabilities from the Monte Carlo propagation and are distinct from the point-estimate cylindrical $R_{\rm peri} < 100\,{\rm pc}$ central-reach counts in Table 15. The closest point-estimate barred candidate is `3673477423666158080` with 5.0 pc. A source may appear twice when it is selected independently by the static and barred seed lists.

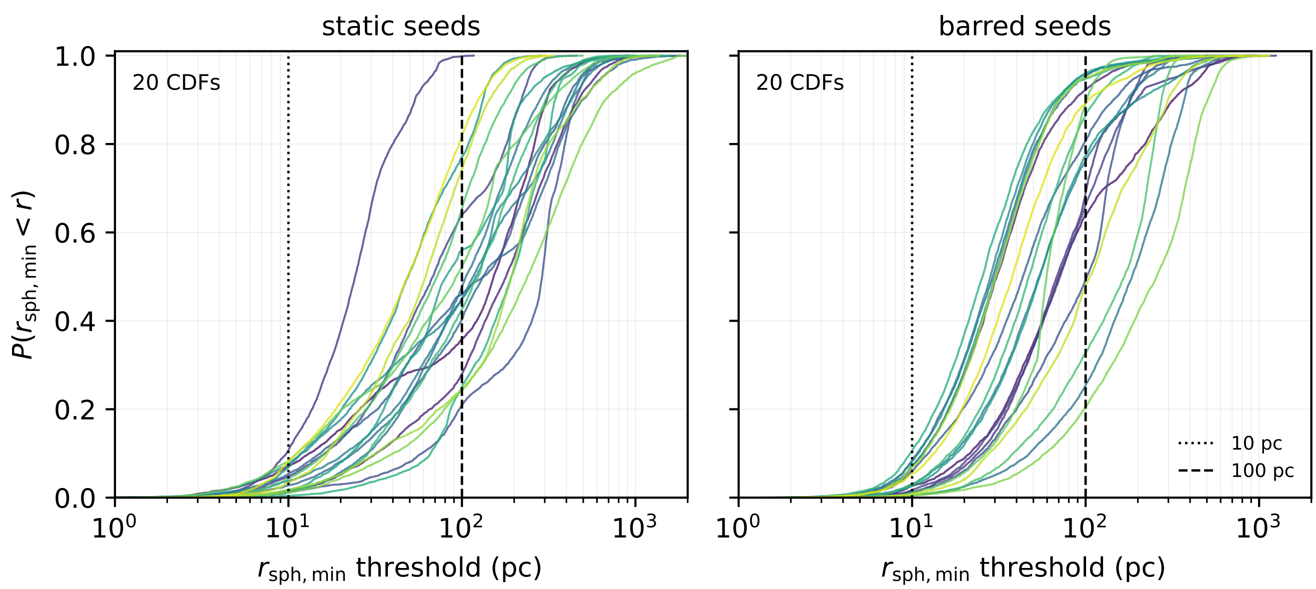

Figure 16. Monte Carlo cumulative probabilities for the 20 closest Sgr A* approachers in the static Hunter+2024 integration (left) and the barred-default integration (right). Each curve uses 5,000 propagated realisations and the same three-point interpolation of the sampled closest approach used in Table 16. The vertical reference lines mark 10 and 100 pc.

4.5. *Robustness and Sensitivity*

The principal dynamical claims are tested here against quality filtering, threshold variation, measurement uncertainty, distance estimator, Solar frame, halo mass, and barred-potential sensitivity.

Gold-subset dynamical robustness in the adopted model.—The Gold dynamical subset starts from the Tier A+B headline catalogue and applies the observational Gold cut before comparing point-estimate orbit summaries. The Gold cut barely changes the orbit distribution: median $R_{\rm peri}$ is 121 pc in the Gold subset versus 120 pc in the full Tier A+B catalogue, and all Gold stars have finite action solutions in the adopted static potential. The subset is intended for individual-star follow-up rather than for redefining the full catalogue.

Table 17. Orbit-derived properties before and after the observational Gold cut. The Gold subset is defined in Table 5; quantities here are point-estimate outputs of the adopted static potential and are not used to define the subset. Eccentricity-fraction errors are Beta$(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011).

Quantity	Tier A+B	Gold obs.
N stars	517	397
Median eccentricity	0.972	0.971
Fraction with $e > 0.8$ (%)	100.0	100.0
Fraction with $e > 0.95$ (%)	$79.9^{+1.7}_{-1.8}$	$78.8^{+2.0}_{-2.1}$
Fraction with $e > 0.99$ (%)	$17.2^{+1.7}_{-1.6}$	$15.9^{+1.9}_{-1.7}$
Median $R_{\rm peri}$ (kpc)	0.1196	0.1210
Median $R_{\rm apo}$ (kpc)	8.20	8.12
Median $z_{\rm max}$ (kpc)	3.92	3.85
Finite action solutions	517	397

Threshold sensitivity within the current local sample.—The $25\,{\rm km\,s^{-1}}$ threshold is a practical operating cut, not a special physical boundary. The candidate pool is therefore released with per-star $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ values and tier labels rather than only a hard threshold flag; alternative threshold cuts can be reconstructed directly from the machine-readable catalogue.

Table 18. Threshold sensitivity. Counts at varying $V_{\rm grf}$ cuts and the corresponding orbital eccentricity distribution under the point-estimate Hunter+2024 orbit integration. The $V_{\rm grf} < 25$ km s^{-1} threshold used as the headline is a practical operating cut, not a sharp physical boundary. Fraction errors are Beta$(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011). Point-estimate fractions are systematically higher than the Monte Carlo posterior-median counterparts in Table 11 because of measurement-error convolution.

Cut (km s^{-1})	$N_{\rm point}$	$N_{\rm ABC}$	$\tilde{e}$	$e > 0.95$ (%)
< 10	169	162	0.987	$99.4^{+0.6}_{-1.1}$
< 15	550	527	0.978	$91.3^{+1.1}_{-1.3}$
< 20	1,337	1,181	0.969	75.8 ± 1.3
< 25	2,591	1,835	0.962	65.7 ± 1.1

Velocity-threshold Monte Carlo propagation.—Adaptive Monte Carlo propagation was run for all 20,829 near-threshold candidates. The baseline pass used 500 realisations per source, with 5,000-realisation refinement in the transition region and 10,000-realisation refinement near the hard $25\,{\rm km\,s^{-1}}$ point-estimate boundary. The resulting tier labels, rather than a single point estimate, define the catalogue.

Table 19. Adaptive Monte Carlo velocity-threshold propagation for the candidate pool. Each star is sampled from the Gaia DR3 parallax–proper-motion covariance, with sky position held fixed, an independent radial-velocity error, and the Bailer-Jones+2021 photogeometric distance posterior. Transition-band stars are refined to 5,000 realisations; near-threshold point estimates are refined to 10,000.

Quantity	Value
Candidate pool	20,829
Tier A+B+C after MC tiering	1,835
Pass-1 realisations / star	500
Transition-band refinements	1,551
Near-threshold ultra refinements	1,197
Refinement realisations / star	5,000
Ultra realisations / star	10,000
Minimum / maximum recovered $P(V_{\rm grf} < 25$ km s$^{-1})$	0.0 / 1.0

Halo-mass sensitivity on the full Tier A+B+C catalogue.—A full-sample halo-mass sensitivity study was repeated for the Tier A+B+C catalogue, bracketing recent Milky Way mass estimates (e.g. Cautun et al. 2020; Wang et al. 2020). Varying the halo mass by ±15% changes the median point-estimate pericentre from 112 pc in the matched 40,001-point default integration to 116 pc and 109 pc, respectively, and leaves the bridger count unchanged at three. The high-eccentricity result is therefore driven primarily by the observed phase-space configuration: all 1,835 Tier A+B+C stars have $e > 0.8$ in the default static potential.

Bar-perturbation sensitivity for a stratified compact-orbit subset.—Because many catalogue stars reach the inner Galaxy, we also integrate a barred Hunter/Sormani variant (Sormani et al. 2022) with pattern-speed sensitivities at $\Omega_p = 24$, 28, 33, 37.5, and $41\,{\rm km\,s^{-1}}$ kpc $^{-1}$. Barred integrations leave the qualitative low-$V_{\rm grf}$ radiality intact but change the central reach and bridger counts. The static potential yields three bridgers, while the barred default yields 22 in the point-estimate integration. The full-sample pattern-speed sensitivity gives 6 bridgers at $\Omega_p = 24$ and $28\,{\rm km\,s^{-1}\,kpc^{-1}}$, 9 at $33\,{\rm km\,s^{-1}\,kpc^{-1}}$, and 24 at $41\,{\rm km\,s^{-1}\,kpc^{-1}}$. The lower end directly tests the slower $\Omega_p \approx 24 \pm 3\,{\rm km\,s^{-1}\,kpc^{-1}}$ bar reported by Horta et al. (2024) from joint APOGEE–Gaia constraints, while the higher end brackets the commonly adopted Milky Way bar pattern-speed range around 35–45 ${\rm km\,s^{-1}\,kpc^{-1}}$

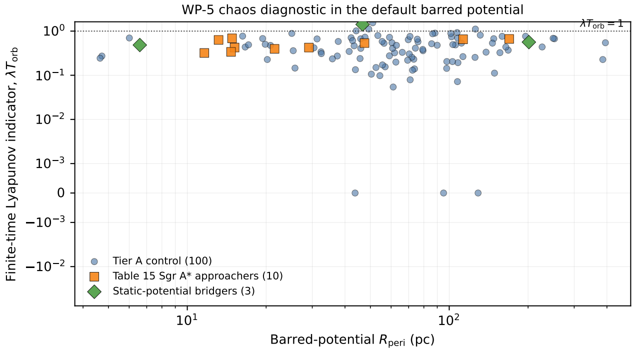

Figure 17. Finite-time Lyapunov indicator in the default barred Hunter/Sormani potential at $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, plotted against the barred-potential cylindrical pericentre. The horizontal dotted line marks $\lambda T_{\rm orb} = 1$, the order-unity level used by Agama as a practical strongly chaotic threshold.

(Clarke et al. 2019; Zhang et al. 2024). At fixed $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, varying the bar angle from 20° to 35° gives 40,001-point median $R_{\rm peri}$ readouts of $\lesssim$ 13–$\lesssim$ 15 pc and $N_{\rm bridge} = 17$–22; flattening the halo over $q_z = 0.80$–1.05 gives median $R_{\rm peri} = 112$–115 pc and leaves $N_{\rm bridge} = 3$. We therefore retain the barred integrations as systematic tests of inner-Galaxy geometry, not as replacements for the static-potential catalogue values. The barred median pericentre readout itself is still sampling-sensitive over 20,001–40,001 points (20 pc to 14 pc in the default barred case), whereas $N(R_{\rm peri} < 100\,\mathrm{pc})$ and the bridger count are stable; we therefore use the barred medians only as resolution-limited diagnostics and rely on threshold counts for robust inner-reach comparisons. This caution is motivated by recent work showing that the Galactic bar can generate or reshape halo phase-space structure, including small-pericentre and chevron-like features in Gaia DR3 (Dillamore et al. 2023, 2024), and that the bar induces measurable chaos in the orbits of nearby halo stars on the very $R_{\rm peri}$ scales sampled by this catalogue (Woudenberg & Helmi 2025). We therefore computed Agama's normalized finite-time Lyapunov indicator, $\lambda T_{\rm orb}$, in the default barred potential at $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ for the three static-potential bridgers, the ten Table 16 Sgr A* approacher seeds, and a random 100-star Tier A control sample. The medians are 0.56, 0.48, and 0.43, respectively; only one of the three bridgers, none of the ten Sgr A* seeds, and four of the 100 controls reach the order-unity regime that Agama flags as strongly chaotic. Figure 17 shows that the closest-approach seeds are not separated from the control sample by the chaos indicator, so the barred central-reach tail is potential-sensitive but not dominated by a uniformly high-chaos subset.

As a second-potential confirmation that the bridger result is not an artefact of the Hunter+2024 disk/halo background, we repeated the Tier A+B+C orbit integration in the made-to-measure inner-Galaxy bar of Portail et al. (2017), constructed via the analytic multi-component Agama recipe of Sormani et al. (2022) and rotated at the matching $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ and 25° bar angle. The Portail+2017 potential uses an independent disk and a flattened Einasto dark halo and therefore samples a different overall gravitational landscape from the Hunter+2024 background. The same 1,835-star catalogue yields a Portail+2017 median $R_{\rm peri} \lesssim 14$ pc in the 40,001-point readout, median $R_{\rm apo} = 8.42$ kpc, median eccentricity 0.996, and $N_{\rm bridge} = 13^{+4.7}_{-3.6}$ (Gehrels 68%); the central-reach counts are also reported in Table 15. This bridger count overlaps the default barred Hunter+2024 value of $22^{+5.7}_{-4.6}$ within Gehrels 1σ, and the median orbital shape statistics have the same high-eccentricity character, demonstrating that the ~tens-of-bridgers headline and the high inner-reach concentration are robust to swapping out the surrounding disc/halo at fixed bar parameterisation.

For the same 100 Tier A control stars, we compared the catalogue Stäckel-fudge actions with actions re-evaluated along the 4 Gyr static-potential trajectories and averaged over time. The median absolute fractional differences are 8.8% in J_r, negligible in J_ϕ ($< 10^{-4}$%), and 81.9% in J_z. The near-exact recovery of J_ϕ is the expected angular-momentum check in the axisymmetric static potential, while the large vertical-action scatter quantifies the Stäckel-fudge caveat for compact plunging trajectories. We therefore keep catalogue actions as follow-up descriptors rather than as selection-defining quantities.

Radial-velocity uncertainty robustness.—As an additional defensive cut, we required Gaia DR3 radial-velocity uncertainty $\sigma_{RV} < 2\,\mathrm{km\,s^{-1}}$ on top of the Gold filter. This removes the noisier RVS solutions and retains 163 of the 397 Gold stars ($41.1^{+2.5}_{-2.4}$%). The comparatively aggressive reduction reflects that DR3 RVS uncertainties are intrinsically larger than $2\,\mathrm{km\,s^{-1}}$ for many of the cool giants that dominate the sample at the typical distance, where the magnitude-dependent inflation of σ_{RV} (Katz et al. 2023) interacts with the $\sim 25\,\mathrm{km\,s^{-1}}$ velocity scale of interest. Within the surviving subset the median eccentricity remains high (0.968 vs. 0.971 for the full Gold sample), the median $V_{\rm grf}$ is similar ($16.57\,\mathrm{km\,s^{-1}}$ vs. $15.72\,\mathrm{km\,s^{-1}}$), and 100% retain $e > 0.8$. The high-eccentricity claim survives this additional quality cut.

Table 20. Galactic-potential sensitivity for the catalogue. Values use point-estimate integrations for all Tier A+B+C stars; all rows use the same 40,001-point, parabola-interpolated trajectory readout. $N_{\rm bridge}$ counts stars with $R_{\rm peri} < 2$ kpc and $R_{\rm apo} > 15$ kpc. Barred median pericentres are resolution-limited readouts at this sampling; bridger and central-reach counts are the stable barred diagnostics.

Variant	$\tilde{e}$	$e > 0.95$ (%)	$\widetilde{R_{\rm peri}}$ (pc)	$\widetilde{R_{\rm apo}}$ (kpc)	$N_{\rm br}$
Hunter+2024 static	0.964	67.5	112	7.43	3
Halo mass 0.85×	0.962	65.3	116	7.44	3
Halo mass 1.15×	0.965	69.2	109	7.44	3
Barred $\Omega_p = 24$	0.996	93.7	$\lesssim 14$	8.00	6
Barred $\Omega_p = 28$	0.996	93.0	$\lesssim 14$	8.00	6
Bar angle sweep	≈ 0.996	91.0–95.4	$\lesssim 13$–$\lesssim 15$	8.30–8.45	17–22
Halo flattening sweep	0.963–0.964	65.9–67.8	112–115	7.43–7.45	3–3
Portail+2017 M2M bar	0.996	94.2	$\lesssim 14$	8.42	13

4.6. *Energy and Angular-Momentum Distribution*

A standard Gaia-era diagnostic is the orbital energy versus z-component of angular momentum (E–L_z); the same diagnostic underlies the substructure surveys of Naidu et al. (2020) and the proto-Galaxy mapping of Rix et al. (2022). In the static Hunter+2024 potential, the Tier A+B+C catalogue occupies low angular momentum and strongly bound inner-Galaxy phase space. This supports the same catalogue-first interpretation used throughout the paper: the sample overlaps familiar Gaia-era accretion and in-situ regions, but the present work does not assign progenitor membership from E–L_z alone. The per-star catalogue reports E, L_z, and Stäckel-fudge actions for follow-up clustering work.

The same L_z axis is the natural place to read off where the catalogue's angular-momentum distribution sits relative to the bar resonances. Figure 19 plots the Tier A+B+C catalogue in L_z–J_R Stäckel-fudge action space and overlays the inner Lindblad, corotation, and outer Lindblad loci derived from the axisymmetrised Hunter+2024 potential for the full pattern-speed grid $\Omega_p \in \{24, 28, 33, 37.5, 41\}\,{\rm km\,s^{-1}\,kpc^{-1}}$. The Tier A points concentrate near $L_z \approx 0$ at large J_R, intersecting the ILR loci of several pattern speeds rather than piling up at any single corotation or OLR; this is the same low-angular-momentum, high-radial-action signature expected of compact-pericentre orbits, and it is consistent with the matched-control OLR-fraction check in Section 4.2.

4.7. *Pipeline Injection-Recovery*

To measure the direct pipeline efficiency, we injected controlled kinematic realisations into GeDR3mock sightlines, photometry, stellar parameters, and Gaia-like uncertainties (Rybizki et al. 2020). The mock contains 10,000 stars in each true-$V_{\rm grf}$ interval 0–25, 25–50, 50–100, and 100–200 ${\rm km\,s^{-1}}$, after the initial $G < 17$, $G_{\rm RVS} < 14$, and RVS-template-temperature cuts. Each injected star is passed through the same Lindegren zero-point, distance-posterior, 500-realisation velocity-threshold Monte Carlo, $\varpi_{\rm corr}/\sigma_\varpi > 5$, and Gaia DR3 RVS-availability stages used by the catalogue. For consistency with the v2 selection-function audit, the RVS-availability draw preserves the v1 GaiaUnlimited convention $g = G_{\rm RVS}$ and $c = BP-RP$. Table 23 and Fig. 20 show that 6,086 of 10,000 true 0–25 ${\rm km\,s^{-1}}$ injections are recovered as Tier A+B+C, while only 282 of 10,000 stars in the adjacent 25–50 ${\rm km\,s^{-1}}$ decade leak into the slow catalogue and the higher-velocity leakage is negligible. Conditional on entering the Monte Carlo after the RVS and parallax-SNR filters, the slow-bin recovery is 6,086/6,944 = 87.6%; end-to-end, including Gaia/RVS observability, it is 60.9%. Combined with the smooth-DF expectation in Section 3.3 and the selection-function prior-dominance audit in Section 5, the injection-recovery test forms the catalogue's completeness triad: the observed slow tail is overabundant relative to a smooth velocity ellipsoid, its effective-volume scaling is limited by sparse selection-function cells, and the pipeline recovers a quantified fraction of true synthetic slow stars.

Table 23. GeDR3mock injection–recovery summary. Each true-$V_{\rm grf}$ bin contains controlled kinematic injections into GeDR3mock sightlines, photometry, stellar parameters, and Gaia-like uncertainties after the initial $G < 17$, $G_{\rm RVS} < 14$, and $3500 < T_{\rm eff} < 7000$ K observability cuts. $N_{\rm RVS}$ is the number passing a stochastic GaiaUnlimited DR3-RVS draw evaluated with the same v1 convention used for the catalogue, $g = G_{\rm RVS}$ and $c = BP-RP$; N_ϖ is the number passing $\varpi_{\rm corr}/\sigma_\varpi > 5$; $N_{\rm input}$ is the intersection entering the 500-realisation velocity-threshold Monte Carlo; and $N_{\rm rec}$ is the number recovered as Tier A, B, or C ($P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.5$).

True $V_{\rm grf}$ bin	$N_{\rm true}$	$N_{\rm RVS}$	N_ϖ	$N_{\rm input}$	$N_{\rm rec}$	$N_{\rm rec}/N_{\rm true}$
0–25	10,000	7,001	9,891	6,944	6,086	0.609
25–50	10,000	7,035	9,896	6,972	282	0.028
50–100	10,000	7,090	9,889	7,030	1	0.0001
100–200	10,000	6,966	9,879	6,906	0	0.000

Table 21. Solar-parameter sensitivity of the Tier-A/B/C catalogue counts and median orbital geometry. The four variants span the literature-quoted range of R_0, $z_\odot$, V_c, and the LSR motion (GRAVITY Collaboration 2022; Schönrich 2012; Reid & Brunthaler 2020). Tier counts vary by $\sim 50\%$ across the variants, but ensemble orbital geometry is stable to $\sim 12\%$ in $R_{\rm peri}$ and $\sim 1\%$ in $R_{\rm apo}$ and eccentricity. Tier counts in this table are computed on the legacy preselection subsample used for the solar-frame re-tiering, so the absolute counts differ from the full-catalogue headline (Tier A/B/C increments 276/241/1318; Table 1); the stable geometry medians are the quantity of interest here.

Variant	R_0 (kpc)	V_c ($\mathrm{km\,s^{-1}}$)	$N_{\rm A}$	$N_{\rm B}$	$N_{\rm C}$	Med. $R_{\rm peri}$ (pc)	Med. $R_{\rm apo}$ (kpc)
default	8.178	232.0	215	116	300	150	8.10
GRAVITY+22	8.275	229.0	234	132	355	154	8.14
LSR-6 (S12)	8.178	232.0	249	156	365	141	7.99
RB20	8.178	248.5	176	100	239	163	8.10

Table 22. Sensitivity battery status for the Tier A+B+C catalogue (N = 1,835). The default static and barred point-estimate orbit products have been regenerated from the parent-complete candidate pool. Halo-mass and bar-pattern-speed rows are point-estimate sensitivity runs using the same initial conditions; the distance-estimator row remains a non-orbit tiering stress test.

Axis	Variant	Med. $R_{\rm peri}$ (pc)	Med. $R_{\rm apo}$ (kpc)	Med. e	N bridgers
Default static	Hunter+2024 axisymmetric	112	7.43	0.964	3
Default barred	Hunter/Sormani barred, $\Omega_p = 37.5$	$\lesssim 14$	8.38	0.996	22
Halo mass	$0.85 \times M_{\rm halo}$	116	7.44	0.962	3
Halo mass	$1.15 \times M_{\rm halo}$	109	7.44	0.965	3
Bar pattern speed	$\Omega_p = 33\ \mathrm{km\,s^{-1}\ kpc^{-1}}$	$\lesssim 15$	8.37	0.996	9
Bar pattern speed	$\Omega_p = 41\ \mathrm{km\,s^{-1}\ kpc^{-1}}$	$\lesssim 13$	8.03	0.996	24
Distance estimator	BJ/parallax coupling stress test	—	—	—	no tier-scale change

Note—The default rows use the higher-resolution orbit products reported in the main orbit table. Halo-mass and bar-speed rows use a uniform 40,001-point, parabola-interpolated sensitivity integration. The barred median pericentre readouts continue to fall between 20,001 and 40,001 points, so they are quoted as resolution-limited diagnostics; the bridger counts and $R_{\rm peri} < 100\,\mathrm{pc}$ counts are the stable barred comparisons.

5. INTERPRETATION AND LIMITS

The strongest statement in this paper is empirical. Gaia DR3 contains a tiered catalogue of stars observed at very low Galactic rest-frame speed: 517 stars at Tier A+B confidence and 1,835 stars at Tier A+B+C confidence. The exact $V_{\rm grf} = 25\,\mathrm{km\,s^{-1}}$ boundary is less secure for individual near-threshold sources, so the catalogue is explicitly probabilistic rather than a single hard-cut list.

A velocity-stratified control comparison across four bands shows that median pericentre grows smoothly with Galactic rest-frame speed in the adopted static potential. This continuous trend shows that the small-pericentre character of the slow-star catalogue is the low-velocity endpoint of a broader kinematic sequence in the observed Gaia DR3 6D catalogue, not a special physical boundary at exactly $25\,\mathrm{km\,s^{-1}}$.

The dwell-time analysis (Section 4.3) and the matched-control gradient (Section 4.2) together make a single physical statement. A kinematic cut at very low $V_{\rm grf}$ selects stars on inner-reaching, radially biased orbits during a low-speed phase. This is related to the same orbital-turning-point mechanism discussed in the accretion-debris literature, but the present catalogue does not identify the selected stars with GS/E or any other progenitor. We emphasise that this strong radial bias is in part a kinematic corollary of the selection itself: a star observed near the Solar radius with $V_{\rm grf} \to 0$ must have near-zero angular momentum and lie close to an orbital turning point, so high eccentricity and small model pericentres are largely expected from the cut. The non-trivial findings are therefore the abundance of such stars relative to a smooth velocity field and their spatial and orbital structure, not the radial bias itself.

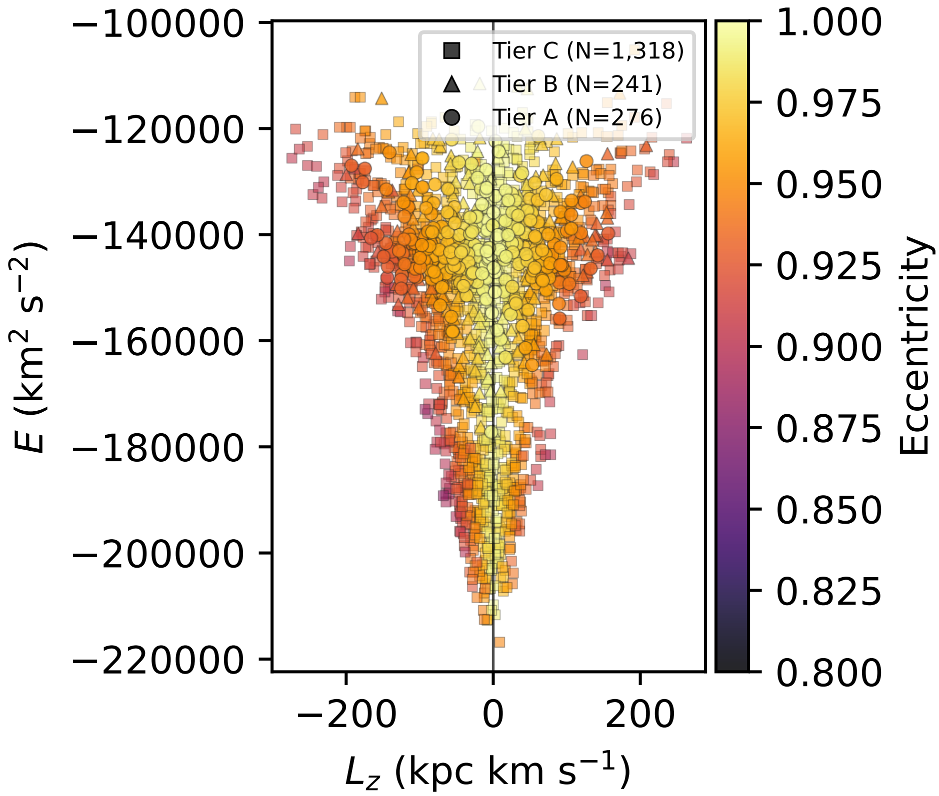

Figure 18. Tier A+B+C slow catalogue (N = 1,835) in E–L_z space, colour-coded by orbital eccentricity in the static Hunter+2024 potential. Marker shape encodes tier (circle/triangle/square = A/B/C); the vertical line marks $L_z = 0$.

The point-estimate orbit integration provides the current characterisation for the main dynamical claim. All 1,835 Tier A+B+C stars have $e > 0.8$ in the default static potential, and $65.7 \pm 1.1\%$ have $e > 0.95$ under point-estimate integration (cf. the $49.2 \pm 1.2\%$ Monte Carlo posterior-median fraction in Section 4.1; the two numbers describe the same sample under different propagation choices). The strongly radial character of the sample is therefore clear in the adopted model, while posterior orbit uncertainties and potential-variant tests remain sensitivity products rather than catalogue-definition steps.

As a final present-day compactness check, we searched the slow catalogue for open-cluster- or association-like aggregates in 3D Galactocentric position (Table 24). No Tier A+B pair lies within 25 pc, and the closest Tier A+B pair is separated by 38.9 pc. The broader Tier A+B+C sample contains three close pairs below 25 pc, but a DBSCAN sensitivity grid (Ester et al. 1996; Schubert et al. 2017) over $\varepsilon = \{5, 10, 15, 25, 50\}$ pc and `min_samples`$= \{3, 4, 5, 7, 10\}$ returns zero compact groups for both Tier A+B and Tier A+B+C. Thus the catalogue is not explained by one or more present-day bound clusters or compact associations. This does not rule out common accretion-related origins or action-space substructure, which need not remain compact in present-day position.

Table 24. DBSCAN present-day compact-group sensitivity grid for the low-$V_{\rm grf}$ catalogue. Entries give the number of compact groups found in 3D Galactocentric position space for Tier A+B / Tier A+B+C.

ε (pc)	`min_samples`				
	3	4	5	7	10
5	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
10	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
15	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
25	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
50	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0

Note. The search uses the Ester et al. (1996) DBSCAN algorithm in present-day 3D Galactocentric position space, with ε varied from 5 to 50 pc and `min_samples` from 3 to 10. The paired entries are Tier A+B / Tier A+B+C group counts. The closest Tier A+B pair is 38.9 pc, while Tier A+B+C contains three pairs below 25 pc but no three-member group at any grid point. This grid follows the parameter-sensitivity guidance of Schubert et al. (2017).

The spectroscopic cross-match sharpens the physical picture while stopping short of an origin claim. The alpha-classified subset contains only 113 Tier A+B+C stars, so population-region fractions are quoted only

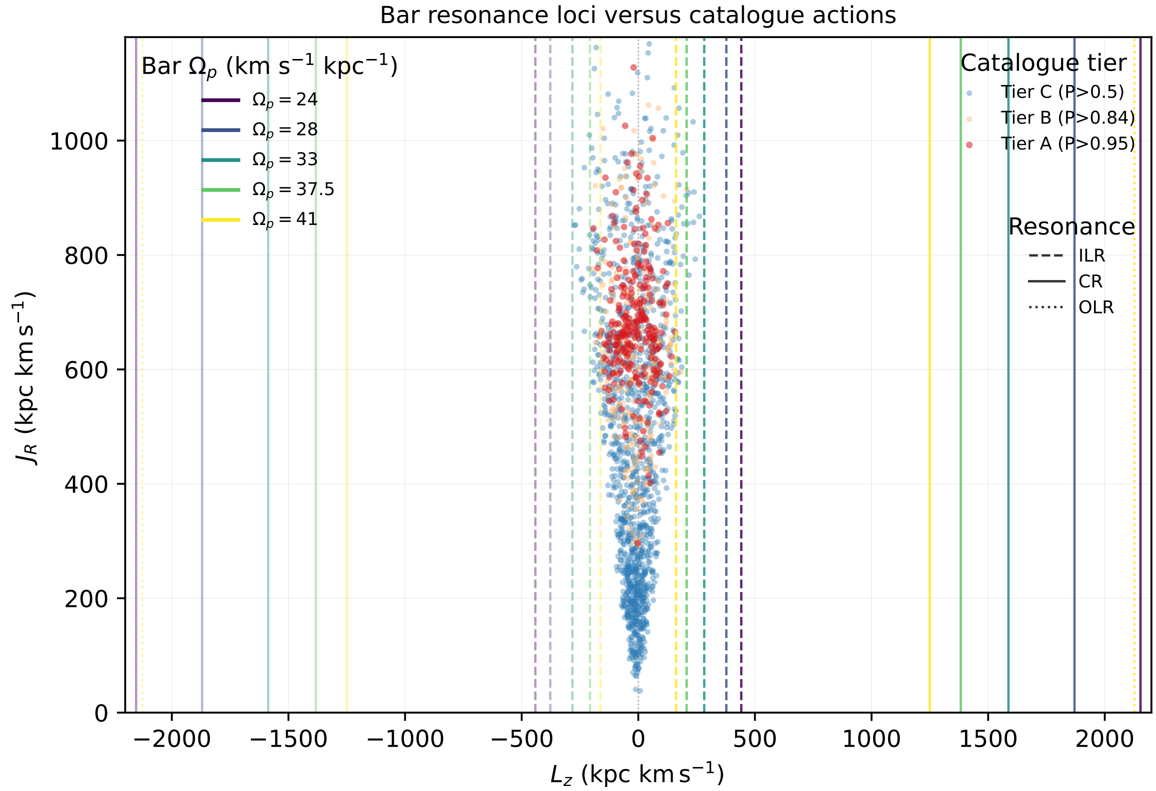

Figure 19. Bar resonance loci versus catalogue actions. Vertical lines mark the values of L_z for circular orbits at the inner Lindblad (dashed), corotation (solid), and outer Lindblad (dotted) radii in the axisymmetrised Hunter+2024 potential at each of the five bar pattern speeds in the extended grid (colour-coded). Loci at negative L_z mirror the retrograde-orbit case. Catalogue points are colour-coded by tier and use the Stäckel-fudge (L_z, J_R) values reported per star; Tier A points cluster near $L_z \approx 0$ at large J_R.

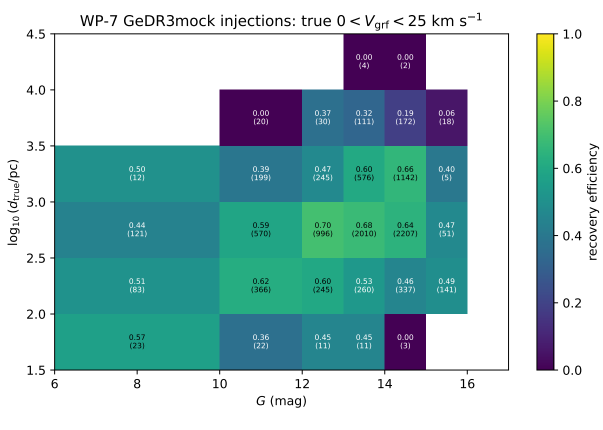

Figure 20. End-to-end recovery efficiency for the true $0 < V_{\rm grf} < 25\,{\rm km\,s^{-1}}$ GeDR3mock injections as a function of apparent G magnitude and true distance. Each cell reports the recovered fraction, with the number of injected stars in parentheses. The denominator is the post-observability injected sample; the numerator requires the RVS-availability draw, $\varpi_{\rm corr}/\sigma_\varpi > 5$, and Tier A+B+C recovery in the 500-realisation velocity-threshold Monte Carlo.

inside that subset. The sample overlaps phase-space regions often discussed in the Gaia-era accretion literature (Belokurov et al. 2018; Helmi et al. 2018; Belokurov et al. 2023), and the high-eccentricity chemistry context has been studied with much larger spectroscopic samples (Lee et al. 2023). This paper is not a membership or merger-decomposition analysis. Because the sample is selected on instantaneous state, not on origin, chemistry, action space, or a fully forward-modelled selection function, we do not attempt progenitor assignment here.

As a selection-function scale diagnostic, we compute $V_{\rm eff} = \sum_i 1/p_{{\rm sel},i}$ over stars with valid GaiaUnlimited cells. For Tier A+B, excluding cells that sit exactly at the GaiaUnlimited prior mean gives $V_{\rm eff} = 396$; including those cells at the prior gives $V_{\rm eff} = 844$, a factor of 2.13 larger. For Tier A+B+C the corresponding values are $V_{\rm eff} =$ 1,392 and 3,168, a factor of 2.28. The difference is therefore best read as a selection-function fallback systematic, not as a volume-complete population normalization.

The prior-dominated (low-parent-count, $n < 10$) fraction can be quantified directly from the underlying GaiaUnlimited (G, $BP-RP$, HEALPix) grid. For Tier A+B, 83.8% of valid stars sit in cells with parent count $n < 10$ and contribute 89.0% of $\sum 1/S_{\rm RVS}$ (Table 9; Fig. 9); restricting to the inner-Galaxy quadrant $l \in [330°, 30°]$ where the catalogue concentrates lowers this to 66.7% of stars and 73.8% of $\sum 1/S_{\rm RVS}$ – still majority prior-dominated. The choice of selection-function estimator is a smaller effect. Combining the Cantat-Gaudin et al. (2023) DR3 parent-completeness map with the Castro-Ginard et al. (2023) RVS subsample SF leaves $\sum 1/S_{\rm RVS}$ for Tier A+B unchanged at 844 to within 0.1% because the DR3 parent completeness saturates at the catalogue's $G_{RVS} < 14$ magnitude range. Substituting the older EDR3 RVS SF of Everall & Boubert (2022) yields a nominally $\sim 18\times$ larger $\sum 1/S$, but that estimator targets a smaller, brighter RVS subsample than DR3 RVS and is not a like-for-like replacement. We therefore retain the prior-dominance audit as the operative SF systematic for the catalogue.

The principal open questions are interpretive rather than catalogue-level. Spectroscopic coverage is incomplete. Very small inferred pericentres and bridger counts are model-sensitive. The sample is not volume-complete, and local halo samples cannot in general be extrapolated to the global halo without strong modelling assumptions (Sharpe et al. 2024). These limitations set the boundary of the present paper's claims.

6. CONCLUSIONS

We present a tiered Gaia DR3 catalogue of stars with very low Galactic rest-frame speeds. The headline Tier A+B catalogue contains 517 stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$; the Tier A high-confidence core contains 276 stars with $P > 0.95$, and the broader Tier A+B+C catalogue contains 1,835 stars with $P > 0.50$.

In the default static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits with median eccentricity 0.949, median $R_{\rm peri} = 151\,{\rm pc}$, and median $R_{\rm apo} = 7.45\,{\rm kpc}$ from the 5,000-realisation orbit Monte Carlo.

The catalogue is not a re-statement of the local velocity ellipsoid. A trivariate-Gaussian smooth-DF expectation matched to the 25–260 km s^{-1} control library predicts 76.5 stars below 25 km s^{-1}; the Tier A+B+C catalogue contains 24.0× that count (Section 3.3, Fig. 6), and Anderson–Darling tests reject the catalogue's eccentricity distribution against both the adjacent 25–50 km s^{-1} matched control and the smooth-DF mock at $p < 0.001$.

The barred-potential sensitivity sweep produces 6–24 bridgers across the pattern-speed grid $\Omega_p \in \{24, 28, 33, 37.5, 41\}\,{\rm km\,s^{-1}\,kpc^{-1}}$, while the static default produces three. An independent M2M bar (Portail et al. 2017) integrated at $\Omega_p = 37.5\,{\rm km\,s^{-1}\,kpc^{-1}}$ yields 13 bridgers and the same compact, high-eccentricity barred-orbit character as the Hunter+24 barred reference; the Gehrels Poisson 1σ intervals for the two bridger counts overlap (Section 4.5, Table 15). Because the barred median pericentre readout is still sampling-sensitive at 40,001 points, the robust comparison is the overlapping bridger count and stable point-estimate cylindrical inner-reach population rather than a precise barred median. This cylindrical $R_{\rm peri}$ count is distinct from the Sgr A* analysis, where the six soft approachers are defined by Monte Carlo posterior probability in spherical Galactocentric distance. The bridger headline is therefore robust to swapping the disc and halo background at fixed bar parameterisation.

A finite-time Lyapunov diagnostic in the barred potential shows 1/3 of the static bridgers, 0/10 Sgr A*-approacher seeds, and 4/100 Tier A control stars in the order-unity strongly chaotic regime (Section 4.5, Fig. 17). Bar-induced chaos is present in this sample but bounded; the orbital shape statistics it underwrites are descriptors rather than selection-defining. A direct check confirms this: comparing Stäckel-fudge actions to time-averaged actions over the 4 Gyr integration gives median absolute fractional differences of 8.8% in J_r, $< 10^{-4}$% in J_ϕ, and 81.9% in J_z.

Selection-function weighting is reported as context, not as a precise deprojection. Quantitatively, 83.8% of the headline catalogue's valid sources sit in GaiaUnlimited cells at parent count $n < 10$, contributing 89.0% of $\sum 1/S_{\rm RVS}$ (Section 3.6, Table 9), and the corresponding effective-volume inflation between the prior-excluded and prior-included sums is 2.13× for Tier A+B and 2.28× for Tier A+B+C. A controlled GeDR3mock injection-recovery exercise (Rybizki et al. 2020) recovers 60.9% of true 0–25 km s^{-1} injections end-to-end and 87.6% conditional on entering the pipeline, with leakage from 25–50 km s^{-1}, 50–100 km s^{-1}, and 100–200 km s^{-1} of 2.82%, 0.01%, and 0% respectively (Section 4.7, Table 23). Smooth-DF excess, prior-dominance fraction, and pipeline recovery together form a three-leg quantitative defence of the catalogue's reality as more than a thresholding artefact.

A DBSCAN sensitivity grid over $\varepsilon \in \{5, 10, 15, 25, 50\}$ pc and `min_samples`$\in \{3, 4, 5, 7, 10\}$

returns zero compact groups for both the Tier A+B and Tier A+B+C samples (Table 24). The catalogue is not explained by one or more present-day bound clusters or compact associations.

Chemistry is subset-qualified. Only 113 Tier A+B+C stars carry the joint [Fe/H] and alpha-element abundance combination required for literature-region comparisons, so Splash, Gaia-Sausage/Enceladus, Aurora, and disc-like fractions are quoted only within that alpha-classified subset.

The natural next catalogue step is Gaia DR4: applying the same probabilistic low-$V_{\rm grf}$ selection to the longer-baseline release will test the stability of the present DR3 sample, retire the prior-dominated (G, $BP{-}RP$) cells that currently set the dominant selection-function systematic, and enlarge the accessible low-speed catalogue.

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AI Assistance Disclosure.—AI language-model tools were used during preparation of this manuscript for coding assistance, editorial revision, literature-navigation support, and critical feedback on draft text. These tools were not used as authors and were not treated as scientific authorities. In particular, no AI tool was used to generate, run, or independently validate the numerical analyses on which the catalogue rests: the Gaia DR3 sample construction, the Monte Carlo uncertainty propagation, the `agama` orbit integrations, the Galactocentric and threshold-membership calculations, the matched-control reweighting, the selection-function diagnostics, the statistical tests, and all figures and tables were produced and verified directly from the documented release-stage pipeline and re-executed by the author. All numerical results, validation steps, methodological decisions, interpretations, references, and final prose were reviewed and verified by the author, who takes sole responsibility for the scientific content.

Data Availability.—The observational source catalogues used in this work are publicly available from the respective survey archives (Gaia DR3 via the ESA Gaia Archive; APOGEE DR17 via the SDSS DR17 archive; GALAH DR3 via the GALAH Survey website; the Queiroz et al. 2023 StarHorse value-added catalogue via VizieR; and the Zhang et al. 2023 Gaia XP parameter catalogue via the GAVO Heidelberg TAP service as `xpparams.main`). The propagated candidate pool, the Tier A, Tier A+B, Tier A+B+C, and Tier D catalogues, the four matched-control band catalogues, the per-star Monte Carlo results, and the orbit-derived columns are archived in Zenodo at https://doi.org/10.5281/zenodo.20116134. The release includes the FITS catalogues, control catalogues, validation products, Gaia Archive selection query, pipeline scripts, potential configuration files, and environment metadata needed to reproduce the numerical products from the released intermediate tables.

Software Availability.—The review-stage analysis pipeline, including sample construction, control-band matching, orbit integration, Monte Carlo propagation, and the figure- and table-generation scripts, is available in a version-controlled GitHub repository at github.com/wodvik/gaia-slow-vgrf-catalogue. The repository also contains the review-stage catalogue products, column documentation, environment file, configuration file, Gaia Archive selection query, and the current manuscript draft.

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